

Quantization fragility under image corruption is recipe-dependent: paired evidence from INT8 and FP8 object detectors

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ABSTRACT

Whether an 8-bit quantized detector stays reliable under corruption cannot be read off the format label: it depends on how the engine was built and which model region is quantized. We pair every corrupted input with a codec-matched clean control, so identical bytes reach executable TensorRT INT8 and FP8 engines, and estimate the corruption-associated change in the FP8–INT8 average-precision (AP) gap, ΔE , inside common-image bootstrap samples. Across nine dataset×scale blocks of YOLO11 on KITTI, VOC, and a 2,000-image COCO subset, ΔE reverses sign across blocks and four blocks survive Holm correction, spanning both positive and negative signs, even though FP8 exceeds INT8 on absolute clean and corrupted AP in every matched block mean. In RetinaNet, a matched-recipe decomposition localizes a severe INT8 level deficit (−21.2 AP on KITTI) to the regression head: leaving that head in floating point restores clean-competitive accuracy without retraining, and a post-hoc operand factorial attributes essentially all of the deficit to activation quantization (weights contribute at most a ~ 0.2 -AP residual), while the same protocol on anchor-free FCOS shows no measurable head-specific deficit. A post-hoc evaluation on the untouched 1,197-image KITTI final holdout reproduces the RetinaNet arm pattern at point-estimate level; the head localization and FCOS null also recur on off-the-shelf COCO-pretrained checkpoints. Corruption-aware calibration transfers inconsistently across two frozen folds, including one significant held-out harm. Robustness should be reported per recipe and per model region.

1. Introduction

Object detectors deployed on edge accelerators increasingly rely on post-training quantization (PTQ) to meet memory, energy, and latency budgets (Krishnamoorthi, 2018; Nagel et al., 2021; Gholami et al., 2021; Jacob et al., 2018; Banner et al., 2019). Two widely used 8-bit formats occupy this setting: INT8, with a mature integer-arithmetic inference path (Jacob et al., 2018; Yao et al., 2021; Wu et al., 2020), and FP8, which trades mantissa precision for exponent range on modern hardware (Micikevicius et al., 2022; Shen et al., 2024). At the same time, deployment images deviate from benchmark conditions through noise, blur, weather, and compression artifacts that measurably degrade deep visual models broadly (Hendrycks and Dietterich, 2019; Agarwal et al., 2024), including object detectors under weather, noise, blur, and compression (Michaelis et al., 2019; Liu et al., 2024). A natural question follows: which quantized format tolerates corruption better? We argue that this question, as usually asked, is ill-posed, because “INT8” and “FP8” name datatypes, whereas an executable detector is determined by a *recipe*: calibration data and preprocessing, quantizer placement and operator coverage, excluded regions, graph transformations, and runtime settings.

Two confounds make naive format comparisons misleading. First, a gap measured only on corrupted images mixes the accuracy difference already present on clean images with the change induced by corruption; when one

format also wins on clean images, the corrupted-input gap overstates the corruption effect. Second, recipes can differ systematically across formats in published pipelines: INT8 engines may be built with calibration distributions, preprocessing contracts, and operator-exclusion rules that differ from their FP8 counterparts—our own historical engines are one such instance (Section 4.2), and the two formats’ different numerical structure invites different recipe choices (van Baalen et al., 2023)—so a “format effect” may in fact be a recipe effect. A credible comparison must therefore pair treatments on identical image bytes, subtract the matched clean difference, and—crucially—vary the recipe deliberately rather than only the datatype.

This paper provides such a study on executable TensorRT engines evaluated end-to-end, with the decoder, non-maximum suppression (NMS), and evaluator held fixed within each detector family. We define a paired estimand ΔE : the change in the FP8–INT8 AP gap between a JPEG quality 95 clean control and each corruption cell, estimated inside common-image bootstrap resamples so that pairing covariance is preserved. Under a frozen analysis contract—the estimands, condition schedule, and bootstrap plan are versioned before the paired analysis, while the YOLO11 engines themselves predate it and are disclosed as such (Section 3.4)—covering YOLO11n/m/x (Jocher and Qiu, 2024) on KITTI, VOC, and a deterministic 2,000-image COCO subset with 14 conditions per block,

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we find that ΔE reverses sign across dataset $\times$ model combinations (Section 4.1). The format label alone does not determine corruption behavior.

The sharpest evidence comes from a second detector family. On RetinaNet (ResNet-50 feature-pyramid-network (FPN) backbone) (Lin et al., 2017; PyTorch Core Team, 2026b), the historical INT8 recipe collapses on KITTI (-42.3 AP relative to a matched FP8 engine); repairing its calibration preprocessing contract still leaves a -21.2 -AP deficit for matched INT8, and a stage-wise recipe decomposition traces that residual to one model region: quantizing only the classification head produces no measurable deficit in the diagnostic arm, quantizing only the regression head reproduces the collapse, and excluding the regression head alone restores clean-competitive INT8 accuracy (49.60 AP, matching FP8-matched at 49.48) without retraining (Section 4.2). The matched collapse is much milder on VOC (-2.9 AP), so the fragile region’s contribution is itself dataset-dependent.

Finally, we turn the diagnostic into a controlled two-fold intervention. If corrupted-input accuracy were limited by calibration on clean images only—ranges that fail to cover corrupted-input activations—then calibrating on corruption-augmented images should restore accuracy on held-out corruption types. Freezing complementary splits (Gaussian noise and JPEG calibrated with fog and motion blur held out, and vice versa) shows that the intervention itself is context-dependent: the frozen held-out criterion is met on KITTI fold 1 ($+1.07$ AP) and VOC fold 2 but fails on the other two cells, including a significant across-the-board harm on KITTI fold 2, and a corruption-specific held-out component is significantly positive in exactly one cell. Three conclusions follow (Section 4.5): excluding the regression head is a validated retraining-free repair of the observed level deficit in the tested recipe; corruption-aware calibration is context-dependent and is not supported as a reliable repair of corrupted-input accuracy; and neither intervention isolates the mechanism behind the head-level failure, which remains open.

This study makes five contributions:

- A paired, codec-controlled evaluation protocol for executable quantized detectors whose estimand isolates the corruption-associated change in a between-treatment gap, with pairing preserved by common-image bootstrap samples.
- Evidence, with a formal heterogeneity test and multiplicity control, that the corruption interaction is context-dependent within one detector family: ΔE changes sign across dataset and model-scale blocks, so the corruption-induced change in the gap has no stable sign on this protocol, while FP8 exceeds INT8 on absolute AP in every matched block mean.
- A recipe decomposition that localizes the RetinaNet INT8 collapse to regression-head quantization, sharpened by a post-hoc operand factorial to the head’s *activation* tensors (INT8 head weights contribute at most a ~ 0.2 -AP residual; the weight $\times$ activation interaction is

compatible with zero on the clean basis and at most 0.1 -AP-scale under corruption), with a non-replication on anchor-free FCOS showing the fragility is not general to INT8 detectors; both the RetinaNet localization and the FCOS boundary recur on off-the-shelf COCO-pretrained checkpoints without evaluation-set checkpoint selection.

- A retraining-free selective-precision INT8 recipe—only the regression head kept in floating point—whose exclusion direction is validated under the frozen analysis contract on both datasets and whose KITTI recovery also appears in a post-hoc point estimate on the untouched $1,197$ -image final holdout ($+30.54$ AP clean); a significant VOC residual remains, and on the audited host the selective engine stays within a few percent of INT8-matched latency and is faster than FP8-matched.
- A controlled corruption-aware calibration intervention over two complementary in-family/held-out folds with codec-matched controls, analyzed with the same paired estimand: the intervention’s held-out effect is itself sign-unstable across fold $\times$ dataset cells, so calibration coverage contributes context-dependently and is not supported as a reliable repair of corrupted-input accuracy.

Section 2 reviews quantization under corruption, detector-specific PTQ, and evaluation methodology. Section 3 defines the protocol, treatments, datasets, and the frozen intervention design. Section 4 reports the interaction landscape, the RetinaNet decomposition and its operand-level factorial, the FCOS second-architecture check, and the intervention outcome. Section 5 discusses mechanism, practical guidance, and limitations. Section 6 concludes.

2. Related work

2.1. Post-training quantization for detectors

PTQ estimates numerical ranges with representative data or data-free surrogates and reduces quantization error through clipping, rounding, or local reconstruction (Krishnamoorthi, 2018; Nagel et al., 2019; Cai et al., 2020; Nagel et al., 2020; Hubara et al., 2021; Li et al., 2021; Wei et al., 2022). Detector-specific PTQ is harder than classification because classification confidence, box localization, feature pyramids, and foreground-background imbalance (Oksuz et al., 2021) must remain jointly calibrated: PD-Quant uses prediction-difference losses, DetPTQ a detection-output loss, and Reg-PTQ specializes calibration and quantization for regression structures (Liu et al., 2023; Niu et al., 2023; Ding et al., 2024). Activation-distribution-aware PTQ likewise addresses activation outliers and non-uniform distributions that strain hardware-friendly per-tensor quantizers (Jiang et al., 2025), and InlierQ suppresses task-irrelevant activation anomalies using a small unlabeled calibration set (Kim et al., 2026). These methods optimize clean-input accuracy; our decomposition instead asks which structural component of a deployed INT8 detector is responsible for fragility, a question Reg-PTQ’s emphasis on

regression structures anticipates but does not measure at the executable-engine level.

Quantization-aware training and distillation approaches (Li et al., 2019; Chen et al., 2021) can produce robust low-precision detectors but require retraining. Our selective-precision arm is deliberately retraining-free: it changes which regions of the graph are quantized, keeping the deployed checkpoint and calibration budget fixed, which makes the recipe contrast interpretable.

2.2. INT8, FP8, and executable treatments

Integer-only inference is an established path (Jacob et al., 2018; Yao et al., 2021; Wu et al., 2020), while FP8 formats now have practical inference and PTQ layouts (Micikevicius et al., 2022; Kuzmin et al., 2022; Shen et al., 2024). Comparisons show that accuracy and hardware efficiency can favor different representations and that the preferred 8-bit format varies by layer and architecture (van Baalen et al., 2023). The nominal datatype therefore does not describe an executable engine: quantizer coverage, scale resolution, excluded operators, calibration, and runtime all contribute. Our protocol records the complete ModelOpt/TensorRT treatment and manipulates recipe components one at a time, which is what allows the recipe-format confound to be measured rather than assumed.

2.3. Corruption robustness and quantized models

Corruption benchmarks (Hendrycks and Dietterich, 2019) measure aggregate degradation for classifiers, and detector studies report task-specific failure under weather, noise, blur, and compression (Michaelis et al., 2019; Liu et al., 2024). The standard summary normalizes mean performance under corruption (mPC) by clean performance (P_{clean}) as the relative rate $rPC = mPC/P_{\text{clean}}$ (Michaelis et al., 2019), which answers a within-model retention question. Our estimand instead asks how the *between-treatment* AP-point gap changes on identical corrupted bytes; a related absolute-AP guard prevents a shrinking gap near a shared failure floor from being read as robustness.

Work linking quantization and distribution shift is comparatively thin: Karimov et al. evaluate quantized detectors under degraded inputs and report sensitivity to the evaluation setting (Karimov et al., 2025), and joint compression-and-robustness designs for detectors treat efficiency and adversarial robustness as coupled objectives (Choi et al., 2026); a separate line of work measures quantization’s effect on *adversarial* robustness (Li et al., 2024), a distinct shift family. TTAQ adapts quantization during continual shift (Xiao et al., 2024), Recti-Q corrects out-of-distribution quantized features (Yaghoubi Araghi et al., 2026), and test-time adaptation more broadly updates the model at inference, including normalization-statistics adaptation under corruption (Wang et al., 2021; Niu et al., 2022). Corruption-aware training and augmentation improve robustness by exposing optimization to degraded inputs (Hendrycks et al., 2020), and detector error analyses show that conclusions depend on how errors are decomposed (Hoiem et al., 2012; Bolya et al.,

2020). Our intervention is related in spirit to corruption-aware calibration, but differs in the experimental control: the calibration corruption families are disjoint from the held-out evaluation families, so any claimed robustness benefit must generalize rather than memorize.

3. Methodology

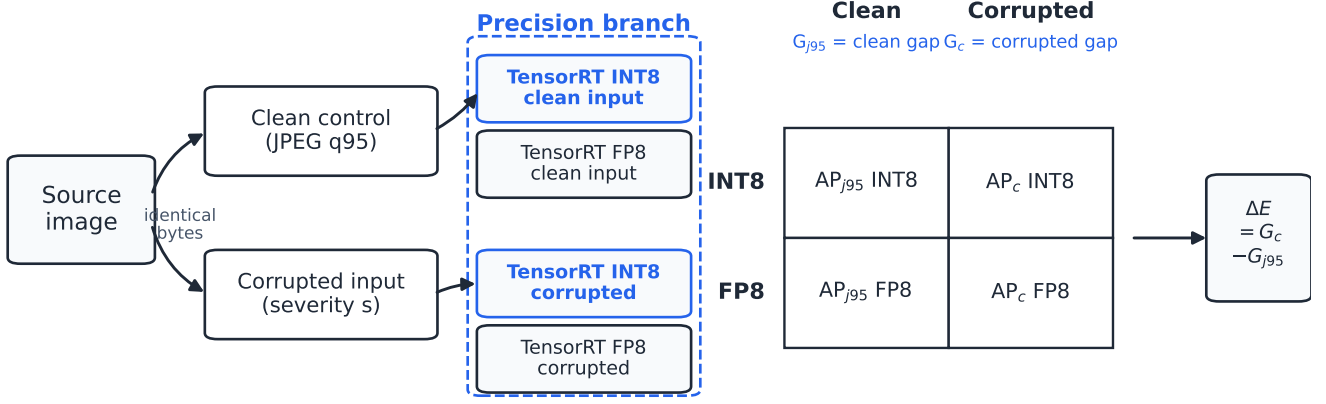
3.1. Executable treatments as recipes

Let a *treatment* denote the complete recorded configuration of an executable detector: checkpoint, Open Neural Network Exchange (ONNX) graph, quantizer placement, quantize/dequantize (Q/DQ) coverage, calibration list, calibration preprocessing, excluded operators, TensorRT builder settings, runtime, and decoder. Throughout, “INT8” and “FP8” qualify such treatments, not isolated datatypes, and an *arm* denotes one treatment instance within an experiment. All engines were built with TensorRT 11.1.0.106 (NVIDIA Corporation, 2026b) on an NVIDIA GeForce RTX 5090 with a 4,096 MiB builder workspace, from ONNX graphs quantized by NVIDIA ModelOpt using entropy calibration (NVIDIA Corporation, 2026a; Migacz, 2017); the study-built registries record ModelOpt 0.45.0, while the pre-study YOLO11 ONNX registries predate that version field.

Builder policy differs across the two evidence programs and is itself a recipe attribute, recorded per engine in the manifests: the YOLO11 engines predate the frozen contract and were built by `trtexec` with TensorRT defaults (TF32 enabled), while the RetinaNet quantization arms built for this study use the TensorRT Python builder with TF32 disabled (`tf32: false` in each engine registry). The RetinaNet FP32 reference and the FP8/INT8-legacy arms are pre-study `trtexec` builds that carry TensorRT defaults, so TF32 is enabled there; within each family the TF32 state is identical across the paired *quantized* arms, which are the only arms entering the ΔE contrasts. Calibration lists contain 512 deterministically selected training images for all paired and intervention arms (the earlier diagnostic sweeps include 128-image and max-estimator variants, marked in the diagnostic table of Section 4.2); evaluation and calibration images are disjoint.

Within this framing we define the RetinaNet recipe arms used below:

- **INT8-legacy:** the historical transferred recipe; its calibration preprocessing does not match inference preprocessing (a contract violation retained as a diagnostic arm).
- **INT8-matched:** entropy INT8 with calibration preprocessing identical to inference (`retinanet_normalized`) and the same 512-image budget as FP8.
- **INT8-selective:** INT8-matched with all `/head/regression_head/` nodes unquantized—under the matched `int8-entropy` build policy those nodes execute in `fp16`—a selective-precision recipe requiring no retraining.



Common-image bootstrap: both precisions are resampled on identical image samples

Accuracy-floor guard: D and corrupted AP support interpretation; they do not transform or censor ΔE

Figure 1: Paired evaluation protocol. A deterministic clean control encoded at JPEG quality 95 (J95) and each corrupted input are created before the precision branch. The four paired cells combine INT8 and FP8 treatments with J95-clean and corrupted inputs; the FP32 engine is a separate reference and cancels from the interaction ΔE . Common image resamples preserve pairing. Clean-control fidelity and absolute corrupted AP—and the signed loss $D = AP_{J95} - AP_c$ shown in the floor guard—are checked before an interaction is interpreted.

- **INT8-corruptcalib** and **INT8-sel-corruptcalib**: the matched and selective recipes rebuilt on a corruption-aware calibration set (noise and JPEG; Section 3.6); **INT8-cc2calib** and **INT8-sel-cc2calib** are the complementary fold (fog and motion blur).
- **INT8-q95calib** and **INT8-sel-q95calib**: codec-matched calibration controls—the same 512 clean images re-encoded at JPEG quality 95 with no corruption (Section 3.6).
- **FP8-matched** and **FP32**: matched-recipe FP8 and the high-precision reference.

One further pre-study engine is cited as context only and is outside this arm list: the **FP8-legacy** exploratory build (a `trtexec-default`, TF32-on engine evaluated under the frozen protocol for reference). INT8-legacy above is likewise pre-study but enters the frozen protocol’s arm table as the documented contract-violation arm. Three senses of “matched” recur and are kept distinct: *contract-matched* arms use calibration preprocessing identical to inference; *codec-matched* inputs share the terminal JPEG quality-95 encoding; and *paired* comparisons evaluate treatments on identical images (and identical resamples in the bootstrap).

Table 1 collects the study terms used throughout.

3.2. Paired estimands and interpretation order

For dataset d , model m , quantized treatment q , and condition c , write $AP_{d,m,q,c}$ for COCO-style AP@[.5:.95] reported in AP points. J95 denotes the clean control: the source image deterministically encoded at JPEG quality 95 with subsampling disabled and no named corruption (Fig. 1). The

Table 1

Terminology used throughout; Sections 3.2–3.4 give the formal definitions.

Term	Meaning
treatment	a complete executable configuration (checkpoint, graph, quantizers, calibration, builder, decoder)
arm	one treatment instance inside an experiment
cell	one corruption $\times$ severity condition (12 per evaluation)
J95 control	clean image re-encoded at JPEG quality 95, matching the corrupted inputs’ terminal codec
level deficit	AP shortfall an arm already carries on clean inputs
ΔE	change in an AP gap under corruption relative to the clean control
severity wall	a saturated cell where all arms sit near the accuracy floor
ledger	a hash-addressed JSON record of an input set, prediction set, or metric
frozen	fixed before the runs it governs; “post-hoc extension” marks analyses added afterward
in-family held-out	/ corruption families present in / absent from a calibration fold

between-treatment gaps on clean and corrupted inputs are

$$G_{d,m,x} = AP_{d,m,FP8,x} - AP_{d,m,INT8,x}, \quad x \in \{J95, c\}, \quad (1)$$

and the primary interaction is

$$\Delta E_{d,m,c} = G_{d,m,c} - G_{d,m,J95}. \quad (2)$$

Positive ΔE means the FP8–INT8 gap widens under corruption relative to the clean baseline; negative means it narrows.

The FP32 reference cancels algebraically. We abbreviate $G_0 = G_{d,m,95}$ and write G_{corr} (“corrupt-mean gap”) for the equal-weighted mean of $G_{d,m,c}$ over the 12 corruption–severity cells; the reported per-block ΔE is $G_{\text{corr}} - G_0$ (Eq. (2)). Absolute corrupted AP is reported beside ΔE ; a shrinking gap near a shared accuracy floor is a floor effect, not robustness (the “floor guard”: interpretations are made jointly with the corrupted AP level). The contrast is descriptive, not causal—algebra analogous to a difference-in-differences interaction (Angrist and Pischke, 2009) without causal identification claims. ΔE is also scale-sensitive by construction: if both treatments retain the same fraction r of their clean-control AP under corruption, then $G_{\text{corr}} = rG_0$ and $\Delta E = (r-1)G_0$, which is negative whenever $G_0 > 0$ and $r < 1$ —a uniform proportional loss shrinks the absolute gap even when neither treatment is anywhere near the accuracy floor. We therefore never read ΔE alone: every reported interaction is accompanied by the absolute corrupted AP and the clean gap, and a negative ΔE means the gap change is context-dependent on the absolute-AP scale, not that no format advantage exists.

The same construction applies to recipe contrasts: substituting two arms of the same format (e.g., INT8-selective minus INT8-matched) isolates a recipe component under an identical protocol, and restricting the corrupted set to a subset S gives the analogue used for the intervention contrasts:

$$\Delta E_{A-B,S} = [\text{AP}_A - \text{AP}_B]_S - [\text{AP}_A - \text{AP}_B]_{J95}, \quad (3)$$

written $\Delta E_{\text{held-out}}$ and $\Delta E_{\text{in-family}}$ for the intervention’s two partitions. The sign is read with the contrast order: for an intervention-minus-baseline contrast, *positive* is a corruption-specific benefit to the intervention arm; under the FP8–INT8 order of Eq. (2), positive instead means the FP8 advantage grows. The sign convention is restated beside each use. This substitution is the key methodological step that turns a format comparison into a recipe experiment.

Two quantities are kept distinct throughout: an arm’s *level deficit*—its AP shortfall to a comparator on a given input set, present already on clean inputs for the RetinaNet INT8 arms (its magnitude equals G_0 when the comparator is the paired FP8 arm)—and the corruption-specific interaction ΔE . The RetinaNet INT8-matched arm exhibits a large level deficit but a negative ΔE on KITTI, so Sections 4–5 keep the two distinct: “level deficit” and “collapse” for the AP shortfall an arm already carries, and “corruption-specific” language (ΔE , interaction) for what corruption adds on top; conflating them would judge the intervention against a target it was never designed to hit.

3.3. Corruption suite and codec control

Four synthetic corruptions are generated deterministically: Gaussian noise ($\sigma \in \{8, 20, 40\}$ on the 0–255 scale), linear motion blur (kernel lengths $\{5, 13, 23\}$ at a per-image uniform angle), synthetic fog (strength–radius pairs (0.18, 6), (0.38, 12), (0.62, 20) blended over a per-image random field), and JPEG compression (qualities $\{70, 40, 15\}$).

Severity labels 1/3/5 order settings within each type. Every output is re-encoded at JPEG quality 95, so the clean control and corrupted inputs share the terminal codec; a SHA-256-derived seed binds dataset, image identifier, corruption, and severity. Within each detector family, inference fixes the input resolution (640-pixel letterbox), confidence threshold, decoder, and non-maximum suppression (NMS): YOLO11 uses a confidence threshold of 0.001 and class-wise NMS at intersection-over-union (IoU) 0.7 with at most 300 detections; RetinaNet uses 0.05 and batched NMS at IoU 0.5, also capped at 300 detections. AP evaluation uses the COCO protocol (pycocotools) with $\text{maxDets}=100$. Decoder constants therefore differ across families, and cross-family comparisons compare recipe bundles rather than a shared decoder; all within-family contrasts share identical decoder settings.

3.4. Datasets and the frozen paired protocol

Three evaluation universes anchor the frozen paired analysis: KITTI validation (1,496 images) (Geiger et al., 2012), PASCAL VOC2007 test (4,952 images) (Everingham et al., 2015), and a deterministic 2,000-image subset of COCO val2017 (Lin et al., 2014) (seed 20260924; the subset is identical across all arms so that COCO comparisons remain paired). VOC and KITTI annotations were converted once to COCO format and frozen; reported values are COCO-style measurements on converted annotations. One checkpoint per dataset×model cell is used throughout, and its provenance is disclosed rather than idealized: the KITTI and VOC YOLO11 scales were fine-tuned from Ultralytics pretrained weights under the pilot program (runs *_train_v1), with the best checkpoint selected on the same validation partition that serves as the evaluation universe here—so KITTI/VOC absolute AP levels carry selection optimism whose residual effect on paired contrasts is unquantified; COCO YOLO11 uses the official pretrained checkpoint with no dataset fine-tune; the RetinaNet ResNet-50-FPN-v2 model was fine-tuned per dataset for 100 epochs at 640-pixel input (seed 20260807) with best-checkpoint selection by minimum validation loss on the same evaluation partition, under the same caveat. The KITTI partition is a frame-level resplit (schedule kitti_confirmatory_v1, seed 20260818) of the 7,481-image official training set into 4,788 train, 1,496 selection (the validation universe reported here), and 1,197 final-holdout images; annotations cover the eight Ultralytics-converted KITTI object categories (car, van, truck, pedestrian, person_sitting, cyclist, tram, misc), with DontCare regions dropped at conversion. It is prospectively locked for this study but not historically unseen (the manifest states this scope note verbatim); frames share driving sequences, so adjacent frames can land on different sides of the split: a perceptual-hash audit (16×16 average hash, per-calibration-image minimum Hamming distance to the 1,496-image evaluation partition) finds 3% of calibration images with an identical-hash evaluation image (13/512; verified visually near-identical at 64×64 mean absolute difference <5/255)

and 12% within distance 5—residual correlation between splits exists and is part of why the bootstrap’s independence caveat in Section 5.4 applies to KITTI. All checkpoint, split, and training-run manifests are hash-bound (Supplement S4). Each dataset×model block evaluates clean, codec-control, and the 12 corruption–severity cells on identical image ordering: 14 conditions per arm, except the RetinaNet FP32 and legacy arms, which predate codec control and carry 13. RetinaNet arms add the intervention engines on the same protocol (Section 3.6).

Two scope notes qualify interpretation. First, the KITTI validation universe was used while developing the selective-precision recipe, so KITTI selective-arm estimates on that partition are development-contaminated; VOC is the cleaner cross-dataset check, and the 1,197-image final holdout—reserved at split time and excluded from all development decisions—provides a post-hoc replication on images excluded from model and recipe development (Section 4.2). This is an image-level, not sequence-level, holdout because adjacent KITTI frames can cross partition boundaries. Second, coverage is asymmetric across families: RetinaNet runs on KITTI and VOC, YOLO11 on all three universes, so COCO evidence is single-family and cross-family comparisons on KITTI/VOC share the same checkpoint-selection caveat.

3.5. Paired image bootstrap

Uncertainty comes from the finite evaluation images with checkpoints, engines, calibration, decoder, and corruptions fixed; evaluation-set composition alone is a known source of score variance (Bouthillier et al., 2021; Pineau et al., 2021). Within each dataset×model block we use one deterministic schedule of 2,000 bootstrap resamples of image indices (Efron, 1979) shared across all arms and all conditions, so every paired contrast is evaluated inside common-image samples and the induced covariance is preserved. Reported point estimates are full-sample plug-in values; reported intervals are 2.5th–97.5th percentiles of the bootstrap distribution of each contrast, marked * when they exclude zero. Two-sided bootstrap p -values use a consistent plus-one convention per tail, $\tilde{p} = \min\{1, 2(b_{\min}+1)/(B+1)\}$ (smallest attainable value $2/2001 \approx 0.001$ at $B=2000$), and are Holm-adjusted across the nine paired YOLO11 blocks (Holm, 1979). Heterogeneity of ΔE across blocks is tested with Cochran’s Q on inverse-variance weights from the bootstrap standard errors (Cochran, 1954; Higgins and Thompson, 2002); because blocks within a dataset share images, their estimates may be correlated; we therefore re-ran the nine YOLO11 blocks under a dataset-shared resample schedule (Supplement) and the observed per-draw correlations of ΔE between model scales are small and mixed-sign (Pearson r spanning -0.20 to $+0.24$, near-zero on VOC and COCO), so the rerun suggests the induced dependence is modest (a generalized Q with a full covariance matrix was not computed). Diagnostic decomposition arms are single-point estimates without intervals; the exploratory layer’s

Table 2

Intervention decision rules and their registration status. “dir.”: the directional reading reported in the text—a significant harm counts as failure. “null-seek”: met only when the interval contains zero (a bound, not an equivalence proof). ^athe registered wording is two-sided (CI excluding 0, either direction); the text reports the directional reading. ^brecorded as a frozen primary estimand, not a success criterion. Codec-matched (q95calib) controls are post hoc for fold 1 and pre-specified anchors in fold 2.

Contrast	Tier	Fold 1	Fold 2
Held-out gain vs clean-calib. arm	full	frozen, dir.	frozen, dir. ^a
Held-out ΔE vs matched/q95	full	post hoc	frozen, dir.
Held-out gain vs clean-calib. arm	sel.	frozen estimand ^b	post hoc
Held-out residual vs FP8 incl. 0	sel.	frozen, null-seek	post hoc

cells carry their own bootstrap intervals in the retained ledger and are reported only as context (Section 4.6).

3.6. Corruption-aware calibration intervention

The intervention tests one hypothesis under decision rules frozen in a versioned configuration before any intervention run: *if the INT8-matched arm’s corrupted-input accuracy is limited by calibration images that never exhibit corrupted-input activation ranges, then corrupting the calibration distribution should recover held-out corruption AP*. The hypothesis is deliberately scoped to corrupted-input accuracy: it cannot by itself account for the clean-image level deficit, which the decomposition attributes elsewhere (Section 4.2). Each of the 512 calibration images is deterministically assigned—by a hash of dataset and image path under schedule nn-corruptcalib-v1—to exactly one of six in-family cells (77–102 images per cell) and corrupted with the same frozen parameters and JPEG quality-95 output encoding as the evaluation caches (Section 3.3). Figure 2 shows the two complementary splits and the codec-matched control: fold 1 calibrates Gaussian noise and JPEG and holds out fog and motion blur, which are never calibrated; fold 2 swaps the roles on the same 512 images (75–95 images per cell). The full tier re-quantizes INT8-matched; the selective tier applies the same calibration to INT8-selective.

Table 2 records the decision rules and when each was registered. The fold-1 configuration carries two success criteria: (i) the full-tier corrupt-calibration arm improves the *held-out* corrupt mean over INT8-matched with an interval excluding zero on the positive side, which would support the range-mismatch hypothesis; and (ii) the selective corrupt-calibration arm’s held-out mean *includes* zero relative to FP8-matched, which would let calibration explain the VOC residual gap of Section 4.2—deliberately null-seeking, since no equivalence margin was pre-specified. The fold-1 primary estimand additionally covers the selective-tier held-out gain over its clean-calibrated counterpart. The complementary fold 2 is frozen in its own configuration before any of its runs and pre-specifies for the full tier (INT8-cc2calib) a replication test of the fold-1 held-out criterion and the corruption-specific rule— $\Delta E_{\text{held-out}}$ of Eq. (3) against both

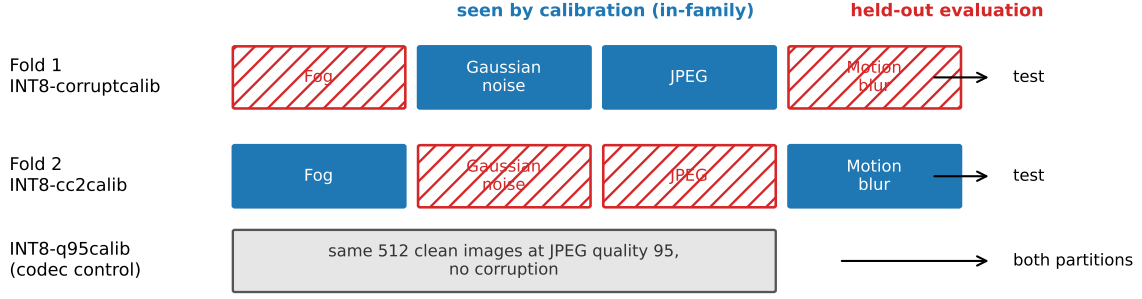


Figure 2: Corruption-calibration folds and the codec-matched control arm. Filled blue cells are families seen by entropy calibration (the in-family partition); hatched red cells are held out and never calibrated. Fold 1 (INT8-corruptcalib) calibrates Gaussian noise and JPEG and is tested on fog and motion blur; fold 2 (INT8-cc2calib) swaps the roles. INT8-q95calib sees only the same 512 clean images re-encoded at JPEG quality 95 and is evaluated on both partitions. Selective-tier arms (sel-corruptcalib, sel-cc2calib, sel-q95calib) share each fold’s split.

the INT8-matched and the INT8-q95calib anchors, excluding zero on the positive side—while its selective-tier contrasts are post hoc.

One wording subtlety matters for how fold-2 outcomes are scored. The fold-2 primary rule is worded non-directionally in the configuration (“CI95 excluding 0 $\Rightarrow$ fold-1 frozen criterion replicated”): the literal registered test detects a held-out change in either direction, and we read “replicated” through the directional fold-1 criterion it cites, so a significantly *negative* held-out change counts as failure. Under the literal two-sided wording, both fold-2 cells would register as “replicated”; the directional reading changes only the KITTI verdict. The directional reading is used throughout; the verdict paragraph reports the literal count as well. In-family improvement alone is evidence of calibration-fitting, not robustness transfer, and all rules are reported regardless of outcome.

Two interpretive analyses accompany the frozen rules and are labeled as such because they were added after the fold-1 freeze (fold 2 subsequently froze the ΔE rule, but not the magnitude comparison): (i) a magnitude comparison of any held-out gain against the residual level deficit to matched FP8, and (ii) the corruption-specific decomposition $\Delta E_{\text{held-out}}$ of Eq. (3), which separates a corruption-specific benefit from a uniform accuracy shift that would appear on clean inputs too. This subtraction rests on an additive, parallel-shift assumption—that any calibration-induced level shift on clean inputs transfers comparably to corrupted inputs—so it is an interpretive decomposition, not a causal identification (Supplement S7).

Two controls discipline interpretation. First, the corrupt-calibration images carry the same JPEG quality 95 terminal encoding as the evaluation caches, whereas the matched clean-calibration list points at original bytes; a gain of the intervention arm over INT8-matched is therefore potentially confounded by codec adaptation rather than corruption coverage. We isolate this component with the INT8-q95calib and INT8-sel-q95calib control arms, calibrated on the same

512 images re-encoded at JPEG quality 95 without corruption. These codec controls were added after the fold-1 freeze and after the fold-1 results were computed, when the confound was identified during review of the intervention design; they are therefore post-hoc controls on the fold-1 contrast and are labeled as such, while fold 2 pre-specifies its ΔE rule against the q95-matched anchor in advance. Second, the codec confound works in the direction that matters for the primary test: a confound that *inflates* the intervention arm’s gains biases the primary superiority test *toward* the positive verdict, so a positive held-out result cannot be attributed to corruption coverage until it survives against the codec-matched anchor. For the null-seeking selective criterion the direction of the confound reverses: codec adaptation that lifts the selective arm moves its negative held-out residual toward zero, so it could in principle manufacture an interval containing zero—a met criterion would be suspect, while an observed failure is conservative.

3.7. Post-hoc extensions: operand factorial, cross-architecture replication, and head-local diagnostics

Two further arms strengthen the localization of Section 4.2 without retraining. *First, an operand-level factorial:* starting from the frozen INT8-matched QDQ graph, we derive two rewired variants of the regression head in which only one operand class is quantized—INT8 weights with fp16-compute activations (W-only) and fp16 weights with INT8 activations (A-only)—by rewiring the head’s quantize/dequantize edges in the ONNX graph while leaving every other node, weight initializer, and calibration output byte-identical to the matched arm. Together with the matched (INT8 weights, INT8 activations) and selective (both operands fp16) arms this gives a 2×2 factorial over head weights $\times$ head activations (interaction estimated as selective–W-only–A-only+matched), so the clean-level deficit can be attributed to weight quantization, activation quantization, or their interaction rather than to the head as a whole. Each rewired graph is validated by diffing every

rewired edge against the corresponding edge treatment in the pre-existing selective-arm graph (fp16 initializer for bypassed weights; direct tensor bypass for dequantized activations), so the arms differ from matched only in which operand the head consumes in quantized form. Rewiring is consumer-local, so “head activations” here means every activation tensor *consumed* by a regression-head node—including the FPN feature maps at the head’s input boundary—not only tensors produced inside the tower; the factorial therefore attributes the deficit to activations in or at the head rather than to internal head activations alone. *Second, a replication on a different detector family:* we train anchor-free FCOS (ResNet-50-FPN) (Tian et al., 2019) on the same KITTI and VOC partitions under the same training profile, export the raw head outputs to ONNX, and run the same matched/selective arm set—the selective exclusion covers FCOS’s regression tower including its centerness output, since torchvision places `bbox_ctrness` inside `head.regression_head`. The FCOS decoder was validated detection-for-detection against torchvision’s reference post-processing on 32 seeded synthetic head-output draws—identical detection sets, zero maximum box and score deviation (Supplement, `decoder_validation.json`; a post-processing math check, not an end-to-end engine parity test); the RetinaNet and FCOS checkpoints (one per detector×dataset) are all early minimum-validation-loss picks (Supplement); the VOC FCOS training loss diverged to NaN at epoch 70, long after its epoch-10 selection point. *Third, an independent-checkpoint replication:* to separate the architecture contrast from our own training pipeline we additionally run the same four-arm set on torchvision’s COCO-pretrained RetinaNet-R50-FPN-v2 and FCOS-R50-FPN checkpoints—no fine-tuning and no checkpoint selection on evaluation data—evaluated on a paired 2,000-image COCO val2017 subset under the same corruption schedule and JPEG-quality-95 codec control, with clean-condition calibration drawn from 512 disjoint COCO train2017 images (calibration manifest hash-bound in the supplement). All three additions are post hoc relative to the paired-protocol freeze: they evaluate frozen conditions under the same paired bootstrap schedule but the arms themselves were fixed after the paired freeze, so we treat them as post-hoc extensions and report them regardless of outcome. A fourth post-hoc diagnostic pair swaps the calibration range estimator itself—INT8-matched and INT8-selective under max calibration instead of entropy, on the same KITTI/VOC graphs, calibration manifests, and 14-condition schedule—to probe whether the head deficit is a tail-clipping artifact (Supplement S14); these are single engine builds reported as plug-in points without intervals. A fifth diagnostic sharpens that counterfactual to the head only: starting from the frozen INT8-matched graphs, the entropy-calibrated scales of the regression-head activation quantizers are replaced by the calibration-set maxima measured on the same captured activations while every other quantizer keeps its frozen scale, and the patched graphs are rebuilt and evaluated on the same 14-condition

schedule (`nn_maxreg_v1_20261001`; single builds, point estimates).

Two further diagnostics measure rather than intervene. Head-activation captures record, on each graph’s 512-image clean-calibration manifest (KITTI/VOC RetinaNet and FCOS, plus COCO-pretrained RetinaNet), every activation tensor consumed by regression- and classification-head nodes of the fp32 graph—including the shared FPN boundary inputs—and summarize each tensor by per-channel $|x|_{\max}$ and excess kurtosis aggregated as medians over channels and tensors, broken down per FPN level; the frozen INT8 graphs’ quantize/dequantize structure is reported in parallel (per-level scale granularity, per-tensor versus per-channel layout). Finally, the four factorial arms (matched, selective, W-only, A-only) are re-evaluated on the untouched 1,197-image KITTI final holdout of Section 3.4—the partition reserved at split time and excluded from all development decisions—on the same 14-condition schedule (`nn_holdout_v1_20261001`; point estimates only, without bootstrap intervals), as a post-hoc check on images excluded from model and recipe development.

3.8. AI-assisted research tooling

An AI coding agent (Cognition Devin) assisted with research-code development and with launching and monitoring the experiment and analysis scripts described above. The authors reviewed every generated or modified script, executed it against the retained manifests and ledgers, and validated the outputs with the repository test suite and the hash-bound evidence package; the agent did not autonomously design experimental conditions, fabricate or manually alter research data, select or omit reported outcomes, or make scientific decisions.

3.9. Evidence layers

Seven evidence layers are kept distinct (Table 3): *diagnostic* evidence (Section 4.2) consists of single-point recipe sweeps that localized the fragile region during development; *frozen paired analysis* is the paired protocol of Section 3.4 with bootstrap intervals—the freeze applies to the evaluation and analysis contract (estimands, condition schedule, bootstrap schedule), not to the artifacts, since the YOLO11 engines, predictions, and evaluation partitions were built for the exploratory program and are shared with the historical layer; *intervention* evidence is the corrupt-calibration experiment and its codec controls (Section 3.6), whose two corruption-calibration folds were frozen prospectively before their engines were built while the fold-1 codec-control arms are post-hoc additions. A fourth, historical exploratory layer—the same paired YOLO11 engines and estimand analyzed earlier as a 144-cell grid over four datasets with a different scope and analysis plan—is reported only as context (Section 4.6). A fifth layer—the operand-factorial and FCOS arms of Section 3.7—runs the frozen paired conditions on post-hoc arms and is reported separately from the frozen content; a sixth, the pretrained-checkpoint replication, applies the same arm set to off-the-shelf COCO-pretrained detectors

Table 3

Evidence layers and their inferential status. Intervals exist only where pairing and the frozen schedule are defined.

Layer	Evidence	Engines/protocol	Uncertainty
Diagnostic	Region-wise recipe sweeps	Development builds	Single-point AP
Frozen paired analysis	Paired 14-cell protocol	Pre-existing engines under a frozen analysis contract	Paired bootstrap (2,000 resamples)
Intervention	Calibration interventions	Frozen fold configs; fold-1 codec controls post hoc	Paired bootstrap (2,000 resamples)
Historical exploratory	144-cell prior grid	Shared engines; separate retrained-holdout and legacy builds	Reported as context
Operand factorial + FCOS	Post-hoc arms on frozen conditions	QDQ-rewired and newly trained engines, unregistered	Paired bootstrap (2,000 resamples)
Pretrained COCO check	Off-the-shelf checkpoints, same arm set	Torchvision RetinaNet/FCOS engines, unregistered	Paired bootstrap (2,000 resamples)
Deployment audits + diagnostics	Max-calibration counterfactual, activation captures, KITTI holdout, rebuild/latency ledgers	Post-hoc point estimates	No intervals

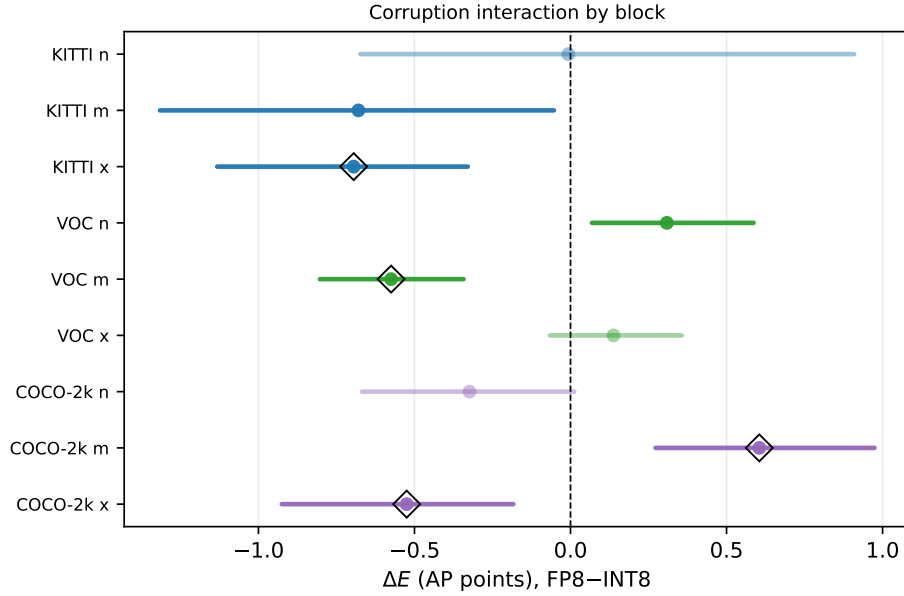


Figure 3: Paired corruption interaction ΔE (FP8-INT8) with 95% common-image-bootstrap intervals. Solid markers denote intervals excluding zero (unadjusted); open diamonds mark blocks surviving Holm adjustment across the nine blocks ($p_{\text{Holm}} < 0.05$). The sign reverses across dataset×model blocks, so the interaction cannot be summarized as a single format-level effect.

on the same frozen COCO-2k evaluation subset, with calibration images drawn disjointly from COCO train2017 (Section 4.4); a seventh layer—deployment-side audits and post-hoc diagnostics (graph-wide and head-local range-estimator counterfactuals, head-activation captures and Q/DQ structure, the KITTI final-holdout re-evaluation, and engine-rebuild and latency ledgers)—reports evidence outside the frozen bootstrap inference; the holdout uses the same AP schedule on a reserved image partition but is reported as point estimates.

4. Results

4.1. The FP8-INT8 interaction is not universal

Table 4 and Fig. 3 report the paired YOLO11 landscape: nine dataset×model blocks, each with the clean-control gap G_0 , the corrupt-mean gap G_{corr} , and the paired interaction ΔE with its 95% interval. FP8 exceeds INT8 on clean-control AP in all nine blocks (G_0 from +0.68 to +3.97 AP points) and on the 12-cell corrupt mean as well (FP8 and INT8 corrupted AP columns, Table 4); the gap is positive on the level in every block, although a minority of individual severity cells reverse it. What lacks a stable sign is the change corruption induces in that gap: it is positive on VOC/YOLO11n (+0.31 [+0.07, +0.59])

and COCO-2k/YOLO11m (+0.60 [+0.27, +0.97]), negative on VOC/YOLO11m (−0.57 [−0.80, −0.34]), KITTI/YOLO11m (−0.68 [−1.32, −0.05]), KITTI/YOLO11x (−0.69 [−1.13, −0.33]), and COCO-2k/YOLO11x (−0.53 [−0.93, −0.18]), and compatible with zero on the remaining three blocks (unadjusted intervals; multiplicity is addressed below). Within VOC alone, the sign of the estimate flips with model scale (n : +0.31 excluding zero; m : −0.57 excluding zero; x : +0.14 null). A claim that INT8 is uniformly less robust under corruption—or uniformly comparable—is not supported on this protocol’s own evidence; within a fixed recipe, the sign of the interaction varies with the dataset×model context.

The heterogeneity is not an artifact of reading nine intervals. Cochran’s test rejects a common ΔE across blocks ($Q = 63.3$ on 8 degrees of freedom, $p \approx 1 \times 10^{-10}$; $I^2 = 87\%$, between-block standard deviation $\tau = 0.44$ AP)—a descriptive heterogeneity summary under an independence approximation—the three blocks within each dataset share the same images, though a dataset-shared-schedule rerun bounds the induced draw correlation at $|r| \leq 0.24$ (Supplement)—so a pooled “format effect” would obscure substantial context dependence. The Holm-adjusted blocks below do not rely on that approximation: Holm requires valid input p -values but no independence among tests, so the cross-block dependence caveat attaches to the pooled and heterogeneity statistics, not to the per-block adjustment. The heterogeneity is not only between datasets: it holds within VOC ($Q = 29.3$, 2 df, $p \approx 4 \times 10^{-7}$) and within the COCO subset ($Q = 21.9$, 2 df, $p \approx 2 \times 10^{-5}$), where only model scale varies, though not within KITTI ($Q = 2.4$, $p = 0.30$). After Holm adjustment across the nine blocks, 4 blocks remain significant and they carry *both* signs—one positive (COCO-2k/YOLO11m) and three negative (KITTI/YOLO11x, VOC/YOLO11m, COCO-2k/YOLO11x; Table 4); a DerSimonian–Laird pooled mean is −0.18 AP ([−0.50, +0.13], a Wald interval)—a descriptive average of heterogeneous interactions, not a common effect. Within one dataset, VOC/YOLO11n and VOC/YOLO11m differ by +0.88 AP ($z = 4.9$ under an independence approximation that is conservative for positively correlated blocks). The heterogeneity also exists *within* blocks at the family level (Fig. 4)—on VOC/YOLO11x the JPEG family contributes −1.07 AP while Gaussian noise and motion blur contribute +0.69 and +0.67 AP—and excluding the saturated motion-blur-s5 cell leaves the cross-block heterogeneity essentially unchanged ($Q \approx 60.2$, $I^2 \approx 87\%$) and every Holm-surviving block’s sign intact, although the two KITTI blocks that do not survive Holm drift (YOLO11n’s null point estimate flips from −0.01 to +0.15 and YOLO11m’s unadjusted-significant negative turns null; Supplement S9, Table S4). The codec control also behaves as required: J95 clean AP tracks original clean AP within 0.3 points for every arm in these nine blocks, and the within-model retention statistic (corrupt mean over J95 clean) ranges 0.68–0.79 across blocks with INT8-versus-FP8 differences of at most

~0.02—the pairing basis itself does not penalize either treatment.

4.2. A matched recipe exposes and localizes the RetinaNet level deficit

The non-YOLO comparator is where recipe effects dominate. Table 5 and Figs. 5–6 show the RetinaNet arms on KITTI and VOC under the frozen protocol. Five observations matter, plus a boundary condition.

First, about half of the historical INT8 recipe’s collapse is a contract violation, not a format property: in the controlled diagnostic—which rebuilds the mismatched contract under the study’s common no-TF32 builder policy at the same 128-image calibration budget and changes only the preprocessing—KITTI J95 AP moves from 6.72 to 27.91, recovering 21.2 of the 41.8-point gap to the diagnostic FP8 build. On the frozen protocol’s original-clean inputs the historical engine reads 7.21 against the matched arm’s 28.29 (Table 5), a 42.3-point total deficit to matched FP8; that pair additionally differs in builder policy (the historical engine predates the study and carries TensorRT’s default TF32 state), so the diagnostic comparison is the attributional one. The legacy recipe’s excess over the matched arm is therefore a mismatched-calibration artifact, not evidence about INT8 per se; what remains after the contract repair is the subject of the next observation.

Second, the residual INT8-matched collapse is severe but dataset-dependent: −21.2 AP versus matched FP8 on KITTI but only −2.9 AP on VOC (original clean images, Table 5; −21.1 and −3.0 on the J95 basis). The paired interaction ΔE (FP8–INT8-matched, corrupt mean) is −4.69 [−5.49, −3.86] on KITTI and −0.26 [−0.58, +0.06] on VOC; the KITTI narrowing is consistent with a scale artifact of the additive estimand: INT8-matched enters corruption 21 points below FP8, and the two arms’ retention rates are nearly equal (0.72 versus 0.75), so a similar relative decline narrows the absolute gap mechanically.

Third, the decomposition localizes the collapse to one model region (Table 6, diagnostic single-point arms on KITTI): quantizing only the classification head preserves accuracy ($50.57 \approx \text{FP32}$), quantizing only the regression head reproduces the collapse (16.41), and INT8 everywhere *except* the regression head recovers to 49.50—within 0.94 of the diagnostic FP32 build on the same J95 basis (50.44; the paired-protocol FP32 reference in Table 5, evaluated on original clean bytes rather than J95, is 50.15) and above the diagnostic full-graph FP8 build (48.50, at the 128-image diagnostic budget). Across all eleven INT8 diagnostic arms the outcome is bimodal in one variable: every arm that quantizes the regression head lands between 6.72 and 28.74 AP, whatever its calibration contract, estimator, budget, or site placement, and every arm that leaves it in floating point lands between 49.14 and 50.57 AP. Two alternatives a reader might raise are inconsistent with this split. Position in the graph is not the explanation, because the classification head is equally terminal yet shows no measurable deficit when

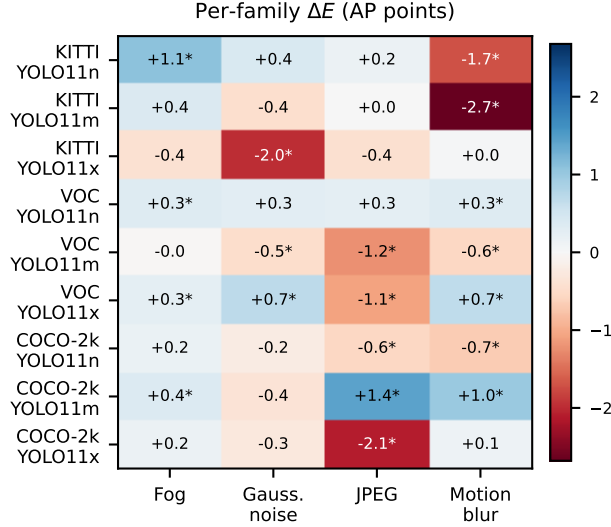


Figure 4: Per-family interaction ΔE (FP8 minus INT8, on the J95-controlled basis) across the nine paired blocks, in AP points. Star: unadjusted 95% bootstrap interval excludes zero. Every corruption family carries both signs across blocks; two blocks are single-signed (VOC/YOLO11n uniformly positive, VOC/YOLO11m uniformly negative), and the remaining seven mix signs—with KITTI/YOLO11x’s one positive cell a near-zero value (+0.03)—the heterogeneity quantified in the text.

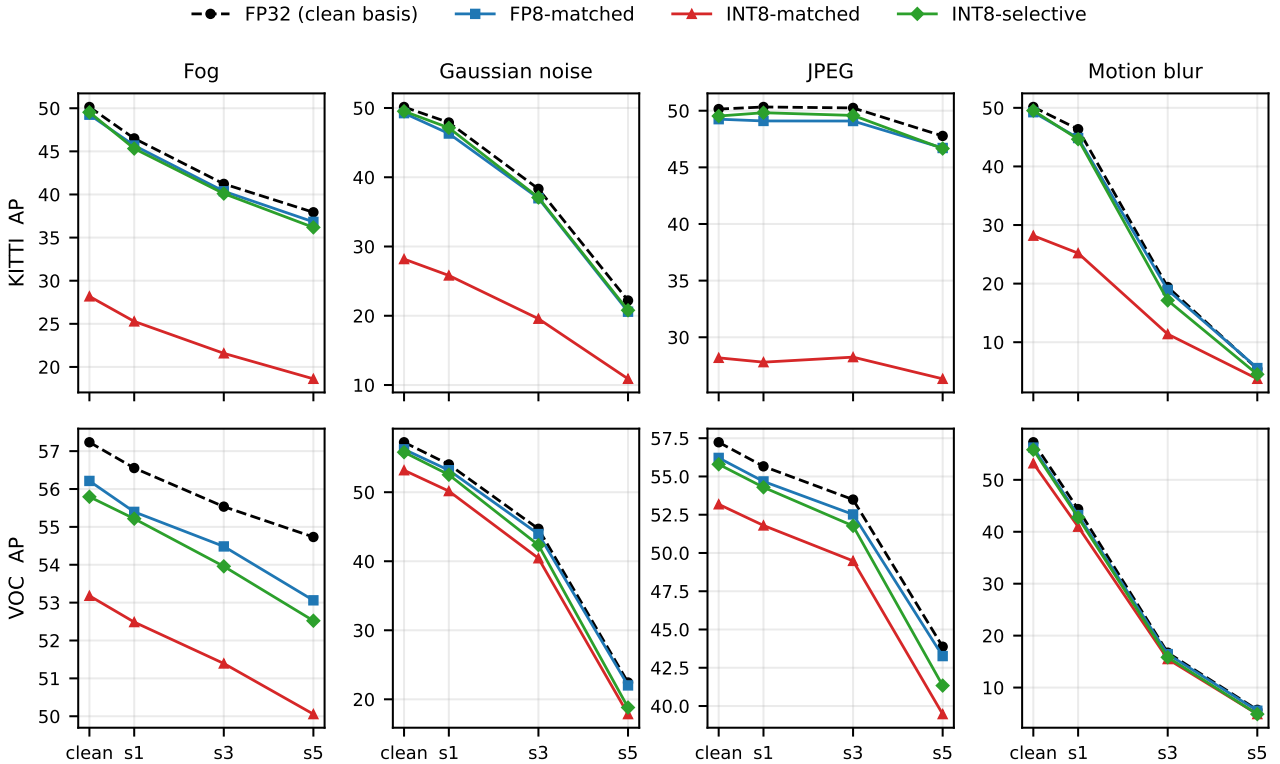


Figure 5: RetinaNet AP by corruption family and severity for the four matched-protocol arms on KITTI (top) and VOC (bottom). The “clean” point is the codec-control J95 cell for the matched arms; the FP32 engine predates codec control, so its baseline is the original-clean value (dashed). INT8-matched sits far below the other arms from the clean baseline onward on KITTI and visibly below them on most VOC cells (the VOC interaction itself is null; Section 4.2), while INT8-selective (regression head in floating point) tracks FP8-matched on most families—its residual widens significantly under motion blur (Section 4.2)—and all four arms converge at the shared motion-blur-s5 failure wall.

Table 4

Frozen paired-analysis contrasts, FP8 minus INT8, per dataset×model block. FP8/INT8 corr. AP: absolute 12-cell corrupt-mean AP (the floor guard); G_0 : clean-control (J95) gap; G_{corr} : corrupt-mean gap (the difference of the two absolute columns, computed before they are rounded to 0.1 AP); $\Delta E = G_{\text{corr}} - G_0$ with paired-bootstrap interval. * interval excludes zero; p_{Holm} : two-sided bootstrap p -value Holm-adjusted across the nine blocks, computed under a consistent plus-one convention $\tilde{p} = \min(1, 2(b_{\min} + 1)/(B + 1))$ at $B = 2000$, so the smallest attainable raw value is $2/2001 \approx 0.001$ and the smallest attainable adjusted value is $9 \times 2/2001 \approx 0.009$; no p is ever exactly zero at finite B .

Dataset	Model	FP8 corr. AP	INT8 corr. AP	G_0	G_{corr}	ΔE	95% interval	p_{Holm}
KITTI	YOLO11n	44.1	41.9	+2.14	+2.13	-0.01	[-0.67, +0.91]	0.915
KITTI	YOLO11m	50.4	47.1	+3.97	+3.29	-0.68*	[-1.32, -0.05]	0.128
KITTI	YOLO11x	51.0	50.3	+1.44	+0.74	-0.69*	[-1.13, -0.33]	0.016
VOC	YOLO11n	44.8	43.8	+0.68	+0.99	+0.31*	[+0.07, +0.59]	0.080
VOC	YOLO11m	51.8	50.7	+1.70	+1.12	-0.57*	[-0.80, -0.34]	0.009
VOC	YOLO11x	54.3	53.3	+0.91	+1.05	+0.14	[-0.07, +0.36]	0.356
COCO-2k	YOLO11n	28.2	27.4	+1.10	+0.78	-0.32	[-0.67, +0.01]	0.171
COCO-2k	YOLO11m	38.0	36.5	+0.89	+1.49	+0.60*	[+0.27, +0.97]	0.016
COCO-2k	YOLO11x	42.4	41.9	+1.00	+0.47	-0.53*	[-0.93, -0.18]	0.024

Table 5

RetinaNet arms under the frozen protocol (AP points, COCO-style AP on converted annotations). Corrupt mean = equal weight over the 12 corruption–severity cells. J95: codec-control clean; FP32 and legacy arms predate codec control (dashes). The INT8-legacy arm is the pre-study historical engine and additionally differs from the matched arms in builder policy (TensorRT-default TF32 state versus TF32 disabled in the TensorRT Python builder); the controlled preprocessing-only contrast is the diagnostic rebuild in Table 6.

Dataset	Arm	Clean AP	J95 AP	Corrupt mean AP
KITTI	FP32 reference	50.15	–	37.80
	FP8-matched	49.48	49.25	36.75
	INT8-legacy (contract mismatch)	7.21	–	7.43
	INT8-matched	28.29	28.18	20.37
	INT8-q95calib (codec control)	28.26	27.85	20.42
	INT8-selective (reg-head fp16)	49.60	49.53	36.58
	INT8-sel-q95calib	50.28	49.79	36.94
	INT8-corruptcalib	29.69	29.74	21.71
	INT8-sel-corruptcalib	49.55	49.03	36.08
	INT8-cc2calib (fold 2)	25.36	25.47	18.68
	INT8-sel-cc2calib (fold 2)	49.72	50.54	37.20
VOC	FP32 reference	57.23	–	42.32
	FP8-matched	56.17	56.21	41.48
	INT8-legacy (contract mismatch)	33.61	–	24.32
	INT8-matched	53.30	53.18	38.71
	INT8-q95calib (codec control)	53.33	53.39	38.54
	INT8-selective (reg-head fp16)	55.87	55.79	40.52
	INT8-sel-q95calib	55.81	55.95	40.40
	INT8-corruptcalib	53.78	53.91	39.35
	INT8-sel-corruptcalib	56.36	56.43	41.13
	INT8-cc2calib (fold 2)	54.15	54.22	39.97
	INT8-sel-cc2calib (fold 2)	56.40	56.43	41.58

quantized alone. Quantization coverage is not the explanation either: the restricted Conv+Add placement (15.75) quantizes fewer sites than full INT8 and does worse, while backbone+FPN quantization with both heads in floating point (49.14) quantizes most of the network and does well. Effects are non-additive (quantizing both heads, 17.09, is worse than quantizing the full graph), so the arms identify a dominant fragile component rather than additive contributions. These arms are single engine builds from one calibration draw on the development set; the frozen paired protocol validates the exclusion direction on both datasets, not the individual diagnostic values.

Fourth—the frozen-protocol check—on the paired protocol the selective arm validates the decomposition: INT8-selective matches FP8-matched within intervals that include zero on the clean control and on the 12-cell corrupt mean on KITTI (FP8 minus selective: J95 gap -0.28 $[-1.11, +0.56]$, corrupt-mean gap $+0.17$ $[-0.07, +0.40]$; we note the J95 interval, -0.28 $[-1.11, +0.56]$, spans over an AP point around zero, so the match is a bound rather than tight equivalence). That match is a statement about the mean, not about every family: measured on the same ΔE basis (family gap minus the J95 gap), the selective residual widens significantly under motion blur ($+1.30$ $[+0.51, +2.04]$) and is null on JPEG (-0.11 $[-0.81, +0.50]$), so family-level divergence remains. On VOC—the cleaner check—a small

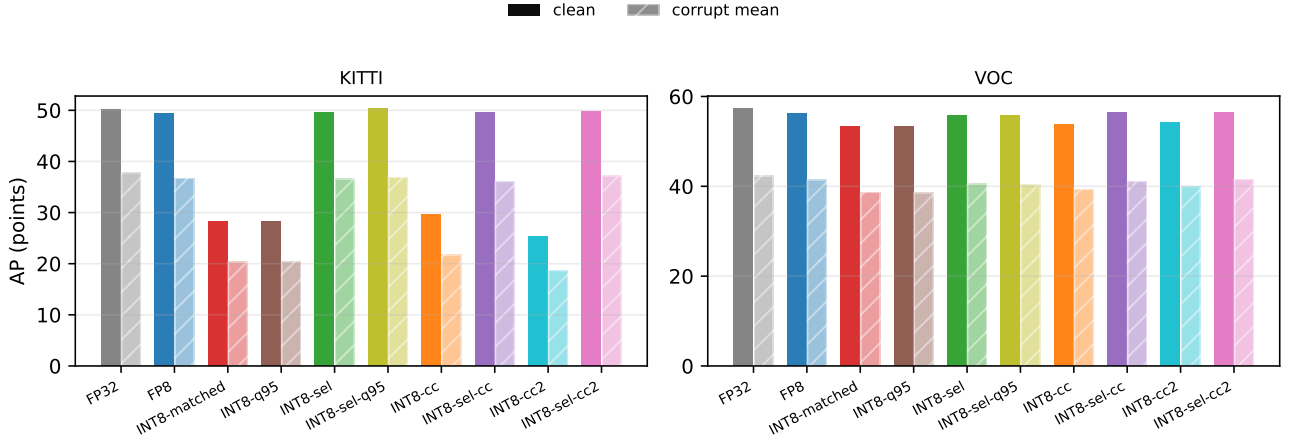


Figure 6: RetinaNet recipe arms on KITTI and VOC: clean AP versus 12-cell corruption mean. INT8-matched collapses on KITTI (dataset-dependent); selective precision (INT8-sel = INT8-selective, reg-head in floating point) restores FP8-level behavior on KITTI and leaves a small residual on VOC; the calibration variants (INT8-q95 = codec-only calibration; INT8-cc = fold-1 corruption-aware calibration on noise and JPEG; INT8-cc2 = fold-2 calibration on fog and motion blur; a “sel-” prefix marks the selective recipe with the regression head in floating point) shift the matched arm little.

Table 6

Diagnostic recipe decomposition on KITTI (single-point AP on the J95 codec-control inputs, 512-image calibration unless stated). These arms guided the paired-protocol design and carry no intervals; the “INT8 matched preprocessing, 512-img calibration” row reproduces the paired INT8-matched cell’s numbers from the same quantized ONNX graph (identical predictions; the serialized engine is a separate build, so the engine hash differs) rather than duplicating it.

Recipe arm (KITTI diagnostic, single-point)	J95 AP
INT8 legacy (mismatched calibration contract, 128-img)	6.72
INT8 matched preprocessing, 128-img calibration	27.91
INT8 matched preprocessing, 512-img calibration	28.18
INT8 max-estimator, 128-img calibration	23.94
INT8 restricted to Conv+Add sites, 128-img calibration	15.75
INT8 at exactly the FP8 compute sites (shared mask)	28.74
INT8 backbone+FPN only (heads fp16)	49.14
INT8 classification head only	50.57
INT8 regression head only	16.41
INT8 heads only (cls+reg)	17.09
INT8 all-except regression head	49.50
FP8 matched, 128-img calibration	48.50
FP32 reference (diagnostic build)	50.44

residual remains on both bases: FP8-matched minus selective is $+0.42$ $[+0.13, +0.74]$ on the J95 clean control and $+0.96$ $[+0.83, +1.09]$ on the corrupt mean, widening under corruption (ΔE $+0.54$ $[+0.23, +0.82]$). The regression head is therefore the dominant contributor, but not the entire mechanism on VOC.

Fifth, the operand factorial (Section 3.7; Fig. 7, Table 7) pins the deficit to a single operand class. Restoring only the head’s *activations* to fp16 while leaving head weights in INT8 recovers the collapse almost exactly: J95 rises to 49.44 AP on KITTI and 55.88 on VOC (100% and 103% of the matched-to-selective gap respectively), and the residual to full head exclusion is sub-0.2-AP in magnitude

(selective minus W-only $+0.09$ $[+0.01, +0.17]$ on KITTI, -0.09 $[-0.20, +0.02]$ on VOC). The complementary arm—fp16 head weights over INT8 head activations—leaves the deficit intact: a clean-level contrast to the matched arm of -0.04 $[-0.20, +0.11]$ on KITTI and $+0.01$ $[-0.08, +0.10]$ on VOC, both intervals enclosing zero, and the same null holds on the interaction basis (ΔE $+0.03$ $[-0.14, +0.20]$ KITTI, -0.03 $[-0.13, +0.06]$ VOC). The head localization is therefore sharper than a region statement: within this recipe the failure is overwhelmingly an *activation*-quantization phenomenon of the regression head—head weights in INT8 contribute at most a residual penalty bounded at roughly 0.2 AP (the W-only contrast above is the weight effect when activations are already fp16, and the A-only contrast is the weight effect when they are not; both are within 0.1 AP of zero in point estimate, though the KITTI selective-minus-W-only interval just excludes it). The factorial interaction is correspondingly compatible with zero on the clean basis ($+0.14$ $[-0.04, +0.30]$ KITTI, -0.09 $[-0.23, +0.05]$ VOC), while the corrupt-mean interaction is a small positive point estimate on KITTI whose interval just excludes zero ($+0.07$ $[+0.01, +0.14]$, unadjusted) and compatible with zero on VOC ($+0.01$ $[-0.03, +0.05]$)—so the two operands act nearly additively, with at most a 0.1-AP-scale interaction on the corrupted inputs, and the weight contribution stays within the bounds already given. This operand attribution is also consistent with why matched FP8 does not fail—its E4M3 exponent range tolerates the head’s activation tails—though this design alone does not separate range clipping from bin resolution on the head’s activation grid (the max-calibration counterfactuals, Supplement S14, point toward resolution rather than calibration-set clipping), and the FP8 comparison quantizes both operands so it does not isolate the activation pathway on its own.

Quantization fragility under corruption is recipe-dependent

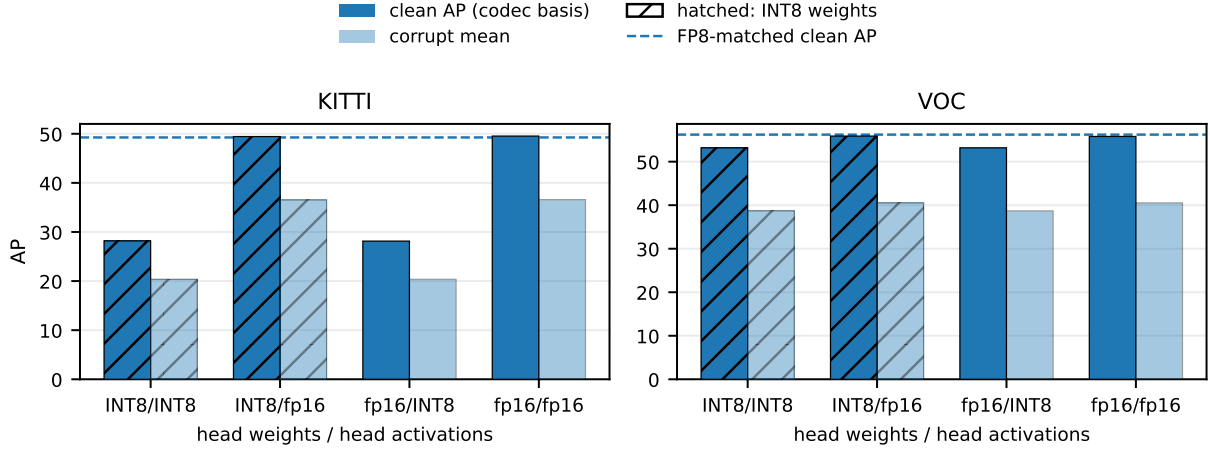


Figure 7: Operand-level factorial inside the RetinaNet regression head (post-hoc arms on the frozen paired conditions). Each group holds head weights and head activations at INT8 or fp16 independently; solid bars are codec-control clean AP and translucent bars the 12-cell corrupt mean. On both datasets the INT8-weights/fp16-activations arm matches the selective arm within 0.1 AP; on KITTI it also reaches the FP8-matched clean level, while on VOC both trail FP8-matched as the selective arm does. The fp16-weights/INT8-activations arm tracks the fully quantized matched arm almost exactly: the deficit follows head *activation* quantization, with a residual weight effect bounded near 0.2 AP on KITTI.

Table 7

The 2x2 factorial of Table 5’s regression-head exclusion, decomposed by operand (post-hoc arms; J95 intervals vs the INT8/INT8 matched arm at 95%, paired bootstrap). “J95 vs W8A8” is the clean-level contrast to the matched arm. Non-quantized operands compute in fp16 under the INT8 engine’s fp16 fallback.

Dataset	Head weights	Head acts.	J95	Corrupt mean	Retention	J95 vs W8A8
KITTI	INT8	INT8	28.18	20.37	0.72	–
	INT8	fp16	49.44	36.52	0.74	+21.25* [+19.88, +22.69]
	fp16	INT8	28.14	20.36	0.72	–0.04 [–0.20, +0.11]
	fp16	fp16	49.53	36.58	0.74	+21.35* [+19.95, +22.81]
VOC	INT8	INT8	53.18	38.71	0.73	–
	INT8	fp16	55.88	40.53	0.73	+2.70* [+2.46, +2.96]
	fp16	INT8	53.19	38.68	0.73	+0.01 [–0.08, +0.10]
	fp16	fp16	55.79	40.52	0.73	+2.61* [+2.37, +2.88]

Finally, a post-hoc KITTI holdout evaluation re-evaluates all four factorial arms on the untouched 1,197-image final partition, reserved at split time and excluded from development decisions (Methods 3.7; same 14-condition schedule; Supplement S15, Table S10). At point-estimate level, without bootstrap intervals, the arm pattern reproduces: INT8-matched scores 34.70 AP clean, while selective reaches 65.25 AP, a gain of 30.54 AP clean and 21.18 AP on the corruption mean. The same factorial structure appears under corruption: W-only is nearly equal to selective on clean and corrupted means (65.30/65.25 and 45.99/46.01 AP), whereas A-only tracks matched (34.64/34.70 clean and 24.86/24.83 corrupt mean), again indicating that restoring head activations rather than weights recovers the deficit. Absolute AP levels are higher on this holdout than on the selection partition for all four arms, but the holdout is image-level rather than sequence-level and adjacent KITTI frames can cross partitions (Section 3.4); the evidence supports replication of the arm structure, not independence from sequence correlation.

A corruption-severity boundary is also shared across precisions (Fig. 5): at motion-blur severity 5 all matched-protocol arms drop to below 6 AP on both datasets—KITTI: FP32 5.45, FP8-matched 5.59, INT8-matched 3.75, selective 4.52; VOC: 5.71/5.54/4.89/4.88 in the same order—FP32 included. That cell measures a common failure wall, not an INT8-specific deficit, and its equal weight in the corrupt mean compresses all contrasts built on it; the leave-one-out sensitivity removes it without changing any frozen-criterion verdict (two secondary codec-control contrasts do move across the zero boundary on VOC; Supplement S9).

4.3. An anchor-free detector bounds the mechanism’s generality

The second-architecture check (Section 3.7; Table 9)—a third detector family in the study, all intervals unadjusted (Section 5.4)—asks whether the activation-fragile regression head is a property of INT8 detectors generally. It does not replicate at a comparable measurable scale: on anchor-free FCOS (Tian et al., 2019) the INT8-matched arm develops no deficit at the RetinaNet scale—J95

Table 8

IoU-threshold and object-size decomposition of the RetinaNet matched-arm deficit on the J95 clean basis (per-cell metric ledgers, single-point values without intervals; all gaps below are measured against FP8-matched).

Dataset	Arm	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
KITTI	FP8-matched	49.2	77.7	52.8	34.0	48.6	61.3
	INT8-matched	28.2	60.3	21.6	3.0	23.2	55.4
	INT8-selective (reg-head fp16)	49.5	77.8	52.9	35.7	48.7	61.7
	W-only (INT8 W / fp16 A)	49.4	78.0	52.7	35.5	48.6	61.6
	A-only (fp16 W / INT8 A)	28.1	60.7	21.6	3.3	23.2	55.4
VOC	FP8-matched	56.2	81.0	60.9	23.6	37.7	62.2
	INT8-matched	53.2	80.2	56.9	17.9	33.8	60.3
	INT8-selective (reg-head fp16)	55.8	80.8	60.5	23.7	38.2	61.5
	W-only (INT8 W / fp16 A)	55.9	80.9	60.7	23.6	38.3	61.6
	A-only (fp16 W / INT8 A)	53.2	80.2	56.9	17.8	33.7	60.3

clean-control AP is 47.53 on KITTI and 49.02 on VOC, against 47.60 and 49.22 for the FP32 reference (12-cell corrupt means 33.82/33.95 KITTI, 34.59/34.88 VOC). Small INT8-to-FP32 gaps do remain: the FP32-minus-selective contrast is compatible with zero on the KITTI clean basis (+0.07 [−0.25, +0.42]) but excludes zero on VOC clean (+0.19 [+0.04, +0.34]) and on both corrupt means (+0.12 [+0.01, +0.22]/+0.30 [+0.26, +0.36]). Those gaps are smaller than the one FCOS’s own FP8-matched arm leaves to FP32 in point estimate (clean 47.60 vs. 47.06 KITTI, 49.22 vs. 48.88 VOC)—and they are not head-specific: head exclusion yields no measurable change (selective minus matched J95 0.00 [−0.15, +0.12] KITTI, +0.01 [−0.04, +0.08] VOC; corrupt mean 0.00 [−0.06, +0.04]/−0.01 [−0.030, +0.005]), so no measurable regression-head INT8 cost is detected for the exclusion to repair. If anything the direction tilts the other way: the matched FP8–INT8 clean contrast is a small negative point estimate on both datasets—compatible with zero on VOC (−0.14 [−0.33, +0.07]) and marginally negative on KITTI (−0.47 [−0.990, −0.001], a borderline unadjusted interval)—and the corrupt-mean contrast is negative on both (−0.50 [−0.66, −0.36] KITTI, −0.08 [−0.15, −0.01] VOC), opposite in sign to RetinaNet’s KITTI matched deficit of >21 AP. The corruption-mean interaction is compatible with zero (FP8-matched minus INT8-matched ΔE −0.04 [−0.44, +0.40] KITTI, +0.07 [−0.13, +0.25] VOC), though—as on RetinaNet—the mean statement does not extend to every family: on VOC the JPEG and motion-blur family ΔE intervals exclude zero. The RetinaNet fragility therefore does not transfer to a differently parametrized head—FCOS regresses per-pixel distances plus a centerness term rather than anchor deltas—which is consistent with recipe-dependence: the collapse tracks a particular head under a particular recipe, not the INT8 format. Two caveats bound the claim: FCOS differs from RetinaNet in decode parametrization, anchor density, and centerness scoring (and its selective arm additionally dequantizes centerness, so the treatments are not identical), so the non-replication speaks to recipe-dependence, not to anchor-based heads as a class; and all four RetinaNet/FCOS checkpoints (one per detector×dataset) are early minimum-validation-loss selections (run-log epochs 21/5 RetinaNet

KITTI/VOC, 14/10 FCOS KITTI/VOC of a 100-epoch schedule; the VOC FCOS training-loss divergence at epoch 70 postdates its epoch-10 selection point and did not affect it); early selection is therefore shared across detectors, but training state is not matched between families, so a detector-specific training-state difference remains a live alternative explanation for the absent deficit.

4.4. Independent replication on pretrained COCO checkpoints

The RetinaNet and FCOS blocks above share one training pipeline, so a shared training-time property could in principle explain both the deficit and its absence. As an independent check we re-ran the four-arm recipe on off-the-shelf Torchvision checkpoints (RetinaNet-R50-FPN-v2 (PyTorch Core Team, 2026b) and FCOS-R50-FPN (PyTorch Core Team, 2026a), COCO-pretrained, no fine-tuning and no checkpoint selection on evaluation data), evaluated on the same frozen paired 2,000-image COCO val2017 subset as the YOLO11 blocks, under the same corruptions and JPEG-quality-95 codec control, with clean-condition calibration on 512 disjoint COCO train2017 images. Table 10 reports the arm levels and the key contrasts.

The direction replicates. On pretrained RetinaNet the INT8-matched arm trails FP32 by +3.60 AP on the clean basis (FP32 38.30, INT8 34.70), the FP8-matched arm sits in between (37.84), and the FP8–INT8 clean contrast is +3.14 [+2.78, +3.57]. Excluding the regression head from quantization recovers most of the gap (selective-minus-INT8 +2.58 [+2.33, +2.85]; FP8-minus-selective +0.56 [+0.22, +0.96]), so roughly 82% of the matched clean deficit is regression-head-borne—the same localization as our trained RetinaNet, at a smaller absolute scale. On pretrained FCOS none of these contrasts excludes zero (FP8–INT8 J95 +0.05 [−0.22, +0.27]; selective-minus-INT8 +0.01 [−0.06, +0.09]), again bounding generality by architecture. Two qualifications apply. First, the pretrained RetinaNet deficit is smaller in absolute terms than our trained-KITTI collapse—consistent with the finding that deficit size is recipe- and context-dependent, and on COCO the FP8-matched arm itself leaves a visible gap to FP32. Second, the corruption-specific interaction ΔE is much smaller than the level gap: on pretrained RetinaNet ΔE is −0.53 [−0.90, −0.14] AP—the FP8 advantage *narrows* rather than widens under corruption, by about a sixth of the level contrast in magnitude—and on pretrained FCOS it is +0.02 [−0.20, +0.28] AP, compatible with zero; so on this larger, cleaner-evaluation base the deficit is primarily a level shift rather than a corruption-amplified one, which bounds how far the interaction framing generalizes beyond the KITTI/VOC blocks.

4.5. Corruption-aware calibration is itself context-dependent

If the INT8-matched arm’s corrupted-input accuracy were limited by clean-only calibration ranges, corrupting the calibration distribution should recover held-out corruption

Table 9

Cross-architecture check (post-hoc arm set): the RetinaNet regression-head deficit does not replicate on FCOS at a comparable measurable scale. Paired-bootstrap contrasts at 95% under the frozen conditions; on FCOS the selective-minus-matched J95 contrast includes zero on both datasets—no measurable head-specific INT8 cost is detected for the exclusion to repair. Detector rows are separate bootstrap blocks; no cross-detector contrast is estimated.

Detector	Dataset	FP8–INT8 J95	FP8–INT8 ΔE	Sel–INT8 J95	Sel–INT8 ΔE	FP8–Sel J95
RetinaNet	KITTI	+21.07* [+19.57, +22.61]	−4.69* [−5.49, −3.86]	+21.35* [+19.95, +22.81]	−5.14* [−5.94, −4.34]	−0.28 [−1.11, +0.56]
FCOS	KITTI	−0.47* [−0.990, −0.001]	−0.04 [−0.44, +0.40]	0.00 [−0.15, +0.12]	0.00 [−0.13, +0.15]	−0.47 [−0.99, +0.05]
RetinaNet	VOC	+3.03* [+2.66, +3.41]	−0.26 [−0.58, +0.06]	+2.61* [+2.37, +2.88]	−0.80* [−0.98, −0.61]	+0.42* [+0.13, +0.74]
FCOS	VOC	−0.14 [−0.33, +0.07]	+0.07 [−0.13, +0.25]	+0.01 [−0.04, +0.08]	−0.03 [−0.09, +0.03]	−0.16 [−0.36, +0.05]

Table 10

Independent pretrained-checkpoint replication on COCO val2017 (paired 2,000-image subset; 95% paired-bootstrap intervals, unadjusted). Arm rows report J95 clean AP and corrupt-mean AP; contrast rows report the named difference on the J95 basis (column 3) and the corruption-interaction ΔE (column 4). RetinaNet reproduces the head-localized INT8 deficit and its selective repair; no measurable FCOS head-specific cost is detected, mirroring Section 4.3.

Detector	Arm / contrast	J95-basis AP	Corrupt-mean / ΔE
RetinaNet	FP32	38.30	31.18
	FP8-matched	37.84	30.58
	INT8-matched	34.70	27.97
	INT8-selective	37.27	29.91
	FP8–INT8	+3.14* [+2.78, +3.57]	−0.53* [−0.90, −0.14]
	Sel–INT8	+2.58* [+2.33, +2.85]	−0.64* [−0.84, −0.42]
	FP8–Sel	+0.56* [+0.22, +0.96]	+0.11 [−0.28, +0.48]
FCOS	FP32	34.32	27.61
	FP8-matched	34.24	27.51
	INT8-matched	34.19	27.44
	INT8-selective	34.20	27.45
	FP8–INT8	+0.05 [−0.22, +0.27]	+0.02 [−0.20, +0.28]
	Sel–INT8	+0.01 [−0.06, +0.09]	0.00 [−0.07, +0.07]
	FP8–Sel	+0.04 [−0.21, +0.24]	+0.02 [−0.19, +0.27]

AP. Table 11 summarizes the four fold×dataset cells on the two bases that must not be conflated: the raw held-out AP change (level basis) and the corruption-specific held-out interaction $\Delta E_{\text{held-out}}$, which subtracts the clean-control shift (Eq. (3)); the full per-arm contrast set is in Supplement S7 (Table S3, Fig. S1).

On the raw level basis, the primary held-out test succeeds on KITTI fold 1 and VOC fold 2, is null on VOC fold 1, and on KITTI fold 2 is a significant *harm* (every level basis degrades, including the families fold 2 calibrated on), so the directional reading fails there; the null-seeking selective-tier test fails on both datasets (held-out residual to FP8-matched excludes zero on VOC −0.51 [−0.66, −0.38] and KITTI −1.30 [−1.61, −1.00]). Even where the primary criterion is met, the KITTI fold-1 held-out gain closes less than a tenth of the residual to FP8-matched on the same cells (−13.34 [−14.13, −12.57] remaining), and excluding the saturated motion-blur-s5 cell changes no verdict (held-out +1.27 [+1.06, +1.50]/+0.07 [−0.10, +0.23]).

On the interaction basis, VOC fold 2 is the only full-tier cell with a significantly *positive* corruption-specific component (+0.62 [+0.30, +0.96], surviving against the codec-matched anchor at +1.06 [+0.76, +1.38]); VOC fold 1

is significantly negative (−0.71 [−1.02, −0.39]), and the two KITTI cells include zero. On the selective tier, KITTI fold 2 lifts the arm where the full tier is harmed (held-out +0.66 [+0.33, +0.92]), but that lift does not survive its codec anchor (+0.25 [−0.09, +0.52]), while the VOC fold-2 selective held-out gain does (+1.50 [+1.35, +1.66], codec-anchored +1.64 [+1.49, +1.80]); the KITTI selective fold-2 arm also exceeds FP8-matched on held-out inputs (+1.04 [+0.64, +1.40], point estimate, under the development-contamination caveat). A retention-scaled sensitivity check (Supplement S7) attributes both KITTI cells largely to scale effects; only VOC fold 2 exceeds its scale-predicted change (observed +1.66 vs. +0.82).

Two codec-matched controls bound a known confound: corrupt-calibration images carry a JPEG quality-95 terminal encoding that clean-calibration images lack, so codec adaptation could inflate any gain (Section 3.6). The quality-95-only arm changes clean AP negligibly (−0.34 [−0.93, +0.35]/+0.20 [−0.07, +0.48]) and does not improve the corrupt mean; level effects stay under half an AP point—far below the 21.2-AP KITTI deficit—while interaction-basis codec effects are the same order as the corruption-specific components, which is why the codec-anchored rows carry the verdict.

Two conclusions follow and should be kept apart. First, on the corrupted-input accuracy the frozen hypothesis actually targets, corruption-aware entropy calibration is itself context-dependent: it produces a significantly positive corruption-specific held-out effect in exactly one of four fold×dataset cells (VOC fold 2), a significant harm on KITTI fold 2, and null or negative effects elsewhere—a second instance of the recipe×context dependence of the paired layer. Second, and separately, neither fold repairs the KITTI level deficit in level terms—but that is a statement about the level deficit, not evidence against range mismatch per se: the design varies calibration image distribution only and never manipulates head-local ranges directly, so the operative mechanism stays open. The contrast that does remove the level deficit is the regression-head exclusion (Section 4.2), a level repair rather than a corruption-specific one.

4.6. Historical exploratory evidence

An earlier 144-cell exploratory grid analyzed the same paired YOLO11 engines—the same estimand ($G_{\text{corr}} - G_{\text{J95}}$), codec control, pairing basis, and KITTI/VOC partitions—on

Table 11

Intervention outcome per fold $\times$ dataset cell, full tier (corrupt-calibration arm minus INT8-matched; AP points with 95% paired intervals, unadjusted). “Held-out change” is the raw AP shift on the held-out families; $\Delta E_{\text{held-out}}$ removes the clean-control shift and is the corruption-specific component.

Cell	Held-out change	$\Delta E_{\text{held-out}}$	Frozen verdict
KITTI fold 1	+1.07 [+0.87, +1.26]	−0.50 [−1.01, +0.03]	primary met
VOC fold 1	+0.02 [−0.13, +0.16]	−0.71 [−1.02, −0.39]	null
KITTI fold 2	−2.02 [−2.31, −1.73]	+0.69 [−0.10, +1.37]	harm (fails)
VOC fold 2	+1.66 [+1.52, +1.83]	+0.62 [+0.30, +0.96]	met + corruption-specific

four datasets including TT100K (Zhu et al., 2016), differing in scope (the full COCO codec-control pool rather than the frozen 2,000-image subset, an extra dataset, and additional arms) and in the analysis plan (no Holm family, different summaries). It found a balanced additive interaction of -0.02 AP $[-0.24, +0.15]$ across the grid and -0.55 $[-0.78, -0.29]$ on post-selection VOC/KITTI holdouts evaluated with independently selected checkpoints, with 56 point-estimate sign reversals after clean adjustment—17 with intervals excluding zero on the negative side. These findings motivated the frozen paired analysis reported above but are not pooled with it. Notably, the historical post-selection holdout estimate is the one piece of this evidence built on checkpoints selected outside the evaluation partition, and its negative sign is consistent with several of the paired blocks. Two exploratory-era transfer cases—RT-DETR-L (Zhao et al., 2024; Carion et al., 2020) and legacy RetinaNet—exhibited catastrophic INT8 clean failures under the unmigrated recipe, evidence in retrospect of recipe fragility rather than format fragility.

5. Discussion

5.1. Recipe and context, not the format label

The evidence layers converge on one conclusion: corruption robustness of a quantized detector is a property of the recipe, the model region, and the evaluation context, jointly with the format. No single layer could show this alone: the frozen paired layer shows the interaction has no stable sign (so pooling it into a format-level constant would obscure substantial context dependence); the diagnostic layer localizes *where* the level deficit lives (the regression head); and the intervention layer shows that the *obvious remedy* (calibrating on corruptions) is itself context-dependent, transferring in one fold $\times$ dataset cell and harming another. Recipe choices—preprocessing contract, quantization sites—move clean AP by 21–42 AP points, several-fold larger than the largest corruption-induced ΔE between FP8-matched and INT8-matched at the 12-cell corrupt-mean granularity (RetinaNet/KITTI: a -4.7 -AP narrowing). Interactions between arms that additionally differ in recipe run higher still—up to $+8.02$ AP on the in-family basis for the INT8-cc2calib–FP8-matched order on KITTI (a narrowing on an arm anchored about 20 AP below FP8, so scale dominates)—and at per-family granularity the FP8–INT8-matched interaction itself reaches -11.4 AP (KITTI motion blur), so the “several-fold”

comparison holds only at corrupt-mean granularity. Reporting “INT8” versus “FP8” robustness without the recipe contract is therefore not merely imprecise; it is capable of reversing conclusions.

5.2. Regression-head localization and possible mechanisms

The decomposition gives a clean localization (diagnostic-level, corroborated by the paired-protocol selective arm): classification-head quantization did not produce a measurable deficit in the diagnostic arm, regression-head quantization reproduces the collapse, and head exclusion restores clean-competitive accuracy. This is consistent with detector PTQ work emphasizing regression structures (Ding et al., 2024). The natural candidate account is RetinaNet’s unbounded anchor-delta parametrization—width/height terms enter an exponential decode, so a fixed additive quantization error on the raw head output becomes a proportional localization error that high-IoU AP penalizes heavily, whereas classification scores pass through a saturating sigmoid and are thresholded. On its own, however, that account does not predict the observed size profile: under anchor-delta coding a fixed relative regression error shifts the decoded box by a factor that is nearly invariant to box size. The metric ledgers show the deficit is anything but size-invariant (Table 8; all gaps versus FP8-matched on the J95 clean basis, single-point values): on KITTI the matched INT8 deficit concentrates at high IoU and small objects (AP₇₅ gap 31.2 AP vs. AP₅₀ 17.4 AP; AP_S collapses to 3.0 vs. 34.0 while AP_L drops only 5.9 AP), and the operand arms reproduce the same profile—A-only tracks the matched arm’s AP₇₅/AP_S almost exactly while W-only and selective match FP8.

A finer-grained reading comes from the activation captures and the quantized graph itself (Supplement S13). The QDQ structure of the matched graph assigns each FPN level its own per-tensor activation scale at the regression head (25 head-consumed quantizers, five shared head-input tensors and four per-level tensors per tower stage), so the deficit is not a single head-wide range being shared across levels; and the captured tails are not uniform across levels but concentrate at the finest pyramid levels—median excess kurtosis 427.5/208.1 at P3/P4 versus 2.8 at P7 on KITTI, against 20.1/19.0 for the classification head, with the same ordering on VOC and on the pretrained COCO graph (Supplement S13, Table S8). Per-tensor grids at the levels that

carry small-object predictions therefore spend the most code budget on outlier coverage, which is the profile under which the AP_S/AP_{75} collapse would concentrate. Two points keep the comparison honest: the head-wide medians pool all head-consumed tensors across levels and tower stages, so they sit far below the fine-level values (head-wide regression kurtosis median 10.6 versus 427.5 at P3 on KITTI), and the two heads differ in tails, not in scale—median per-channel $|x|_{\max}$ is comparable (8.6 versus 10.3 on KITTI) while the head-wide excess-kurtosis medians differ (10.6 versus 3.1). This is a distribution-level consistency argument, not causal isolation: the captures measure the calibration distribution, and the per-level scales already adapt to it.

The intervention adds the sharper negative: on KITTI, where the collapse lives, corruption-inclusive calibration does not repair the head and produces no corruption-specific held-out benefit in either fold—the second fold harms every level basis—so the failure is consistent with a region-level quantization-error sensitivity that none of the tested calibration distributions repairs. A post-hoc range-estimator counterfactual bounds the estimator choice: a graph-wide swap of the entropy estimator for max calibration, which spans the observed calibration range and so removes the entropy estimator’s clipping of it, does not repair the deficit but deepens it (14.02 versus 28.29 AP on the KITTI clean basis) while the head-exclusion benefit persists under it (9.47 AP selective-minus-matched clean), and on VOC the max estimator collapses the classifier’s score distribution outright (J95 clean $\approx$ 0.001 AP on both arms; Supplement S14). Because that swap is graph-wide it perturbs every quantizer, not only the head’s, so it bounds rather than isolates the head-local estimator choice. A head-local counterfactual closes that gap: applying the max estimator to the regression-head activation quantizers only, with every other quantizer byte-identical to the entropy-matched graph, leaves VOC functional and still does not repair the deficit on either dataset (27.15/52.32 AP clean versus 28.29/53.30 entropy-matched; Supplement S14). The head sensitivity is therefore not an entropy-calibrator artifact, and the clipping-versus-resolution reading is more consistent with a coarse-grid (resolution) explanation than with calibration-set clipping: widening the range to the calibration maximum does not recover accuracy. Max calibration does not guarantee that evaluation-time activations stay in range, so clipping of out-of-range evaluation activations is not excluded, and resolution is not further separated from the rounding bias a coarser scale induces. The small corruption-specific transfer observed on VOC fold 2 shows calibration coverage is a real but unstable contributor, which the context-dependent framing already predicts. FP8, which quantizes the same head with an exponent field and therefore a wide dynamic range, holds 49.19 AP (the pre-study FP8-legacy engine) and 49.48 AP (FP8-matched) on KITTI clean inputs, which also fits a dynamic-range account. The operand factorial sharpens this to the activation pathway: within the head, INT8 weights with fp16 activations recover 100%/103% of the matched-to-selective gain (KITTI/VOC) while fp16

weights over INT8 activations differ from the fully quantized arm by approximately zero, with intervals spanning zero (-0.04 [$-0.20, +0.11$] KITTI, $+0.01$ [$-0.08, +0.10$] VOC clean-level contrasts). INT8 storage of the head weights carries at most a residual penalty under ~ 0.2 AP (detectable on KITTI only once activations are restored). The cross-architecture check is at least consistent with a head-shape account: FCOS’s regression head emits per-pixel distances decoded linearly through an anchor-size scaling—without the exponential anchor-delta decode—and develops no head-specific INT8 deficit under the same recipe (Section 4.3, Table 9). That consistency is weak evidence, not confirmation: FCOS also differs in anchor density and centerness scoring, its selective arm dequantizes an extra output—and the exponential decode in the account above runs after the engine, downstream of the activations the factorial implicates, so the mechanism remains at the operand level. Activation captures sharpen but do not resolve this comparison: FCOS regression-head median excess kurtosis at P3/P4 is 14.4/11.3 on KITTI and 3.8/4.7 on VOC, far below the corresponding RetinaNet values (427.5/208.1 and 179.1/143.0). The KITTI FCOS P3 value remains positive, so its tails are lighter, not absent; together with the architectural differences, this is distribution-level consistency rather than causal isolation (Supplement S13, Table S8). Why the head-exclusion effect is about 8 $\times$ larger on KITTI than on VOC (21.3 versus 2.6 AP on the J95 basis) remains open; the per-level tail profile above is a candidate moderator, since small-object prevalence differs across the two datasets.

5.3. Practical guidance

For deployment recipes, the evidence supports a concrete checklist: (i) verify the calibration preprocessing contract byte-for-byte against inference before interpreting any quantization gap; (ii) under matched contracts, evaluate the regression head and classification head as separate quantization regions, in the spirit of sensitivity-based mixed-precision assignment (Dong et al., 2019; Huang et al., 2025)—head exclusion recovered > 20 AP on KITTI RetinaNet while promoting only the `/head/regression_head/` subgraph to floating point, but bought nothing measurable on FCOS (selective minus matched J95 0.00 [$-0.15, +0.12$]/ $+0.01$ [$-0.04, +0.08$]), so the split is worth evaluating, not assuming; (iii) do not rely on corruption-augmented calibration to transfer: across two folds its held-out effect ranged from a significant harm to a $+1.66$ -AP gain depending on the dataset $\times$ family partition, and codec-match the calibration input encoding to the deployment stream before attributing gains to corruption coverage; (iv) report paired interactions together with absolute corrupted AP and the clean gap, so floor effects cannot masquerade as robustness. Selective precision is a retraining-free intervention worth evaluating first: it matched FP8-matched within intervals on KITTI and trailed it by $+0.42$ J95-clean / $+0.96$ corrupt AP on VOC. On deployment cost, an idle-gated single-host audit (CUDA-event timing, three repetitions of 500 timed batch-1

iterations; Supplement S12) measures the selective repair at 1.3–6.4% above INT8-matched latency and INT8-matched engines at 1.16–1.31 ms per image versus 4.62–5.09 ms for the FP32 references that exist—a modest premium for a large AP recovery, on one host rather than a hardware guarantee. Against the natural deployment alternative, FP8-matched engines run at 1.35–1.49 ms, so the selective INT8 engine is 8–11% faster than FP8-matched on every audited detector–dataset pair. Engine-build determinism is also now audited: fivefold rebuilds of the seven RetinaNet arms (KITTI/VOC/COCO matched and selective plus the KITTI A-only variant; YOLO11 engines were not rebuilt) are bit-distinct in every case (35 builds) yet produce identical measured AP on the evaluated cells. The repair is not free in principle: the promoted subgraph covers the regression-head nodes, which a layerwise MAC count of the torchvision graph (analysis/regression_head_macs.py) puts at 24.5–25.1% of convolution MACs at the frozen input size for the two reported class counts. All remaining quantization sites stay INT8-configured in the recipe; kernel-level execution rates were not measured. The W-only variant—head weights keep their INT8-quantized representation in the ONNX graph while head activations are promoted to fp16—differs from full head exclusion by approximately zero on VOC and within 0.2 AP of it on KITTI, so keeping head weights quantized loses nothing measurable beyond that bound (fp16 activation compute is still required, as under selective exclusion; we verified the quantized representation at the graph level, not serialized-engine storage or kernel-level memory).

5.4. Limitations

Artifact and engine scope. The evidence is conditional on recorded artifacts: one checkpoint per dataset–family cell, one engine build per arm (a fivefold rebuild audit of seven RetinaNet arms—the KITTI/VOC/COCO matched and selective INT8 engines plus the KITTI A-only arm; YOLO11 engines were not rebuilt—found every rebuild bit-distinct at engine and inspector level, with identical measured AP on the evaluated cells; Supplement S12), one fixed calibration manifest per arm (a single 512-image draw; resampling variance of the calibration draw is unmeasured), one calibration method (entropy), and toolchain (ModelOpt/TensorRT) on the paired and intervention arms—the diagnostic sweeps additionally vary the estimator and the 128/512-image budget, and a post-hoc max-calibration pair tests the range estimator at the frozen 512 budget (Supplement S14)—and one runtime environment (TensorRT 11.1, RTX 5090); other PTQ methods may place quantization error differently. The mechanism analysis covers one architecture family deeply (RetinaNet, anchor-based) with a cross-family check on one anchor-free detector (FCOS), where no comparable measurable deficit is detected; RT-DETR engines exist only under the legacy recipe and were excluded from the paired protocol, and the original head-region decomposition arms are diagnostic single points without intervals (the post-hoc operand-factorial and FCOS arms do carry paired intervals).

Uncertainty coverage. Bootstrap intervals cover image resampling only—not training-seed, calibration-draw, or corruption-realization variance (engine-build spread is separately audited: fivefold bit-distinct rebuilds, identical measured AP; Supplement S12)—and treat images as independent; KITTI frames drawn from shared driving sequences are correlated—a perceptual-hash audit finds 3% of the 512 calibration images with an identical-hash near-duplicate in the 1,496-image evaluation partition (Section 3.4)—so intervals there may be optimistic, and a sequence-clustered bootstrap is not computable from the retained manifests because drive metadata is not preserved in the evaluation ledger. Holm adjustment covers the nine paired YOLO11 blocks; RetinaNet, intervention, operand-factorial, and FCOS intervals are per contrast and unadjusted. Within the FCOS extension’s 56 contrast cells, 12 intervals exclude zero. A Holm correction across those 56 cells is unattainable at the current bootstrap resolution: the smallest attainable two-sided p under the plus-one convention is $2/2001 \approx 0.001$, which already exceeds the first-step threshold $0.05/56 \approx 9 \times 10^{-4}$, so no FCOS cell can claim adjusted significance from these caches (raising B would be required before any adjusted claim). FCOS statements in the text are therefore unadjusted interval statements only, and we flag them as such rather than mixing bootstrap evidence with a normal approximation for the same claim. The null-seeking selective-versus-FP8 criterion bounds the residual but specifies no equivalence margin, so an interval containing zero is not a proof of equivalence (Lakens, 2017).

Provenance and protocol scope. Three artifact-provenance caveats bound the paired layer: the KITTI and VOC evaluation partitions coincide with the checkpoint-selection partitions for all three detector families (and the COCO YOLO11 models use the official pretrained checkpoint), so absolute AP levels are optimistic; both paired arms share the selected checkpoint, but whether the two arms’ selection biases differ—and hence how much enters the gap and the interaction—is unquantified; and the YOLO11 engines predate the frozen contract and carry TensorRT’s default TF32 state, while the RetinaNet quantization arms were built with TF32 disabled (the RetinaNet FP32 reference and legacy arms are TF32-on trtexec builds, but the ΔE contrasts only join quantized arms built under the same policy), so the families also differ in builder policy (FCOS is internally TF32-off; Supplement). The paired protocol’s freeze covers the analysis contract (estimands, conditions, bootstrap schedule), not the artifacts it runs on—the nine-block Holm adjustment is an analysis-time choice implemented in the reporting code rather than a line in the frozen configuration. Historical exploratory results share the paired YOLO11 engines; only their retrained-holdout and legacy-transfer arms use different builds, and all are reported only as context.

Corruption suite and intervention scope. The corruption suite is synthetic and four types wide; severities order settings within a type and are not comparable across types, and the corrupt mean’s equal weighting includes the saturated motion-blur-s5 wall cell (reported alongside a leave-one-out sensitivity). The intervention uses two complementary folds over these types; other partitions and non-synthetic shifts are untested, and each calibration cell saw only 75–102 images. The intervention arms also differ on the clean control (the corrupt-calibration gains include a significant uniform clean-AP shift), which is why the corruption-specific reading relies on the held-out ΔE decomposition rather than the raw held-out gain.

Registration and anchoring caveats. The fold-1 frozen criteria were a held-out significance test and a selective-versus-FP8 residual test—the magnitude comparison and held-out ΔE analysis followed as interpretive—while fold 2 was frozen after that analysis and pre-specifies both rules, and the codec controls attached to the fold-1 contrasts are post-hoc additions. The fold partition also confounds corruption family with spectral character (fold 1 calibrates the high-frequency families and holds out the low-frequency ones), “held-out JPEG” is not fully unseen in codec terms because every calibration image is already JPEG quality-95 encoded, and the many intervention contrasts carry no familywise multiplicity correction, so individual significant cells should be read against the full table rather than in isolation. On the RetinaNet arms the J95 clean control departs from original clean by up to 0.8 AP (the fold-2 selective arm), larger than the ~ 0.3 -point fidelity seen on YOLO11—enough to move small selective-tier residuals across significance boundaries on the original-clean basis (retained in the ledger). KITTI’s selective-arm numbers are development-contaminated on the selection partition—mitigated by the untouched-holdout replication (Section 4.2), which reproduces the arm structure rather than the absolute levels—VOC serves as the cleaner check, and the two-dataset scope limits the localization’s generality.

Evaluation scope. COCO evidence is restricted to the frozen 2,000-image subset shared by the YOLO11 and pretrained-replication blocks, and AP is a single utility summary; downstream risk thresholds belong to the application. The pretrained replication uses official torchvision checkpoints—removing checkpoint-selection coupling to our training pipeline—but at one model scale (ResNet-50) and one dataset, so it validates the direction and localization of the RetinaNet deficit rather than its magnitude across recipes.

6. Conclusion

A paired, codec-controlled protocol with a common-image bootstrap turns “is INT8 or FP8 more robust to corruption” into a question with a defensible answer: the interaction ΔE changes sign across datasets and model scales—there is no stable format-level winner—while

FP8 exceeds INT8 on absolute clean and corrupted AP in every matched YOLO11 block mean. In RetinaNet the severe INT8 level deficit localizes to the regression head, and leaving that head in floating point is a validated retraining-free repair (+21.3 AP on the J95 basis on KITTI, residual to FP8-matched -0.28 $[-1.11, +0.56]$), with a post-hoc clean-gain point estimate of +30.54 AP on the untouched 1,197-image KITTI final holdout, which was excluded from development decisions; a post-hoc operand factorial attributes essentially all of it to activation quantization, with INT8 head weights contributing at most a ~ 0.2 -AP residual—an attribution that also replicates on the holdout. Anchor-free FCOS shows no measurable head-specific deficit under the same protocol, and the head localization and the FCOS null recur on off-the-shelf COCO-pretrained checkpoints, so the collapse tracks a particular head under a particular recipe rather than the INT8 format. The mechanism picture is consistent but not closed: RetinaNet regression-head activations are heavy-tailed with the tail burden concentrated at the fine FPN levels, whereas FCOS shows substantially lower fine-level kurtosis despite having no comparable head-specific deficit; max-calibration counterfactuals—graph-wide and head-local—are more consistent with a coarse activation grid than with calibration-set clipping, though resolution, rounding bias, and evaluation-time clipping are not separated. Corruption-aware calibration transfers inconsistently across two frozen folds—including one significant held-out harm—so calibration coverage is a secondary, unstable contributor rather than a reliable repair. Robustness should be evaluated and reported per recipe—with paired inputs, clean-gap adjustment, and absolute corrupted AP—rather than per format label.

Data and code availability

The Supplementary Information restates the full frozen contract: estimands, condition schedule, arm registry, inference/evaluation constants, corruption-aware calibration schedule, bootstrap schedule, evidence layers, and a reporting checklist. Manuscript tables and figures are generated deterministically from frozen ledgers and manifests. The immutable reproducibility package accompanying this submission is release v3.0.3 (version DOI <https://doi.org/10.5281/zenodo.23082850>) under concept DOI <https://doi.org/10.5281/zenodo.22031663>. Its SHA-256 manifest binds the packaged commit and retained evidence. The release includes the revised manuscript and supplementary PDFs, analysis code, generated tables and figures, activation captures for FCOS, and the post-hoc KITTI holdout metric, input, run, and engine-registry records. These ledgers regenerate the reported tables and points. Raw datasets, trained checkpoints, raw predictions, ONNX graphs, and built TensorRT engines remain outside the redistribution boundary; re-running inference requires the documented build pipeline.

Declaration of competing interest

The authors declare no competing interests.

CRedit authorship contribution statement

Dinh Thuan Nguyen: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing—original draft. **Lam Phuong Nguyen:** Validation, Data curation, Writing—review and editing. **Vinh Huy Nguyen:** Methodology, Visualization, Writing—review and editing. **Sy Vu Quang:** Validation, Writing—review and editing. **Mohan Rajesh Elara:** Supervision, Writing—review and editing. **Anh Vu Le:** Supervision, Project administration, Writing—review and editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Cognition Devin for manuscript drafting support, consistency checking, and language editing and, as described in Section 3.8, for research-code development and experiment monitoring; OpenAI Codex/ChatGPT assisted literature discovery, manuscript critique, and language editing; Anthropic Claude Code assisted manuscript critique and language editing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

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